Incidence and Impact of Housing Subsidies in Urban Slums: Evidence from India

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Abstract

Improving housing conditions in urban slums is a key challenge in developing countries. Public programs to transition slum dwellers into formal houses require them to demolish their existing dwelling and make complementary investments — raising concerns that the poorest households will be excluded. This paper studies India's flagship house construction grant, PMAY, and finds that despite these participation costs the program is progressive: the share of households receiving the grant decreases by about a fifth (4.6pp) from the 5th to the 95th percentile of the income distribution. The key mechanism is that the value of the existing dwelling is an income-correlated participation cost — richer households have more to give up and are less likely to take up. Eligibility for the program leads to 12pp higher grant take up and 0.09sd increase in house quality index. The improved house provides better protection from extreme rainfall and heat. A welfare framework shows that the program take-up rate is a sufficient statistic for welfare benefits, and that holding the budget fixed, giving a smaller grant to more households would increase aggregate welfare transferred.

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1 Introduction

Over a billion people across the world live in slums with poor quality shelters and unhygienic conditions.[1] Governments in low-income countries have designed housing programs to move them into better houses in the same neighborhood or elsewhere. However, households often need to supplement government subsidies with private investments, or overcome ordeals like application costs and arranging alternative accommodation during construction, or suffer disruption of economic networks. This raises the concern that the poorest households get excluded from such programs. Further, because typically such programs raise the quality of house to a fixed level, the net improvement in house quality would mechanically be lower for people with a better house to start with. The incidence and impact of housing programs is therefore ambiguous and depends on the program's design.

In this paper I study Pradhan Mantri Awas Yojana (PMAY), the flagship housing scheme of the Government of India costing 1-2% of the national budget and targeted at 20 million poor urban households. In this *in situ* housing program beneficiaries receive Rs 200,000 (USD 2,500)[2] in installments contingent on verified construction milestones to construct a house with a concrete roof on the same land that they currently occupy. The beneficiaries need to demolish their existing structure to lay a new foundation that can support a formal house. Households need to have tenure documents for the land to receive the benefits.

To identify the causal impact of the program I exploit variation from a parallel program, the Government of Odisha's JAGA Mission. The government gave non-transferable tenure documents to those occupying state or municipal lands (henceforth *tenable* households) but lacked jurisdiction to do so for those occupying private, temple, railway, forest, or central government lands (henceforth *untenable* households). Since slum dwellers settle on vacant lands with no knowledge of its formal owner, this generates exogenous variation within the slum in the eligibility for housing grant.

The government conducted a census of all slum dwellers that serves as a baseline measure of household income and house quality. I use administrative land records data to identify the formal owners of the land. I conducted two rounds of endline surveys among 6,001 slum households across 11 towns, six years (short-run) and nine years (medium-run) after the roll out of this intervention, to measure the incidence and outcomes.

Despite the ordeals of demolishing their house, arranging alternative accommodation, and mak-

[1] United Nations Sustainable Development Goals Report, 2023

[2] This is more than the median annual household income

ing complementary investments upfront, the incidence of the program is progressive even among slum dwellers[3]. The share of households who have received the benefits decreases by 4.6pp from the 5th to the 95th percentile of the income distribution (with a mean of 21%). The progressivity is driven in part by the richer households opting out: conditional on being invited to apply, self-reported application rates decrease by 4.4pp from the 5th to the 95th percentile, self-reported willingness to take up the program decreases by 10pp, and the threshold level of benefits at which they would take up increases by 16% from the 5th to the 95th percentile households.

The key mechanism is the opportunity cost of taking up the program: the value of the house they must demolish is higher for richer households who have typically invested more in their current dwelling. Since house quality is positively correlated with their income, this generates progressive take-up. The income gradients in stated take-up fall by about a quarter when we control for baseline house quality.

Tenable households have a house quality index 0.09σ (p-value: 0.01) higher than untenable households in the same slum nine years since the start of the program. This is a large effect considering the small first stage effect on receiving the grant of 12pp. The impact on the index is driven by a 9.6pp ITT effect on having a concrete roof. Other measures of house quality which are not monitored in the program do not improve. Comparing changes in house quality since 2017, households that received tenure documents but not the grant improved no more than households that received neither, while those that also received the grant improved by about 1σ.

The improvement in house quality better protects the households from extreme weather. Their house is less likely to leak during rain and is cooler on the inside by 1.9℃(LATE, significant at 10%) on hotter days. This enables them to better adapt to climate change.

I find small and insignificant effects on multiple health, education, income, and consumption measures. The households have a 28% higher outstanding debt (not statistically significant) likely taken to fund the house construction. Effects on income and consumption may be lower due to the loan repayment even in the medium run.

Given these results, a central question in the design of such programs is whether, for a fixed budget, the government should give a larger grant to fewer households or a smaller grant to more. A larger grant means a bigger transfer to each recipient, but fewer recipients, and it draws richer households with better existing houses into the program. The welfare-maximizing grant size therefore depends on which households select in. To study this, I develop a framework in the spirit

[3]The program would be even more progressive if we considered the entire city population

of the Roy model in which a household takes up the grant only when its utility gain exceeds the opportunity cost of demolishing its existing house, up to an idiosyncratic preference shock. The key result is a sufficient-statistic characterization: under an extreme-value assumption on the shock, the average welfare households derive from the program depends only on their take-up probabilities and a dispersion parameter, independent of the functional form of utility.

In a stated-preference elicitation conducted during the endline households stated whether they would take up the grant at benefit levels from Rs 50,000 to Rs 500,000. Combining these data with the framework, I show that at the current grant size, a smaller grant distributed to more households would raise the average welfare transferred—both by reaching more households and by screening out richer households with the least to gain.

This paper contributes to multiple literatures. First, it contributes to the literature on ordeals and the incidence of public programs. Ordeals such as application costs can screen out households who value a benefit least (Nichols and Zeckhauser, 1982; Alatas et al., 2016). But ordeals can make incidence regressive when they are harder for the poor to overcome, because of application costs (Romero and Singh, 2023; Gupta, 2026; Finkelstein and Notowidigdo, 2019; Deshpande and Li, 2019) or behavioral frictions such as inattention (Shepard and Wagner, 2022; Bertrand et al., 2004). Models of take-up typically assume that participation costs do not vary with income (Kleven and Kopczuk, 2011). I show that when participation requires forfeiting an existing asset, the asset's value is a participation cost that rises with income, so incidence is progressive despite the ordeal. This is the logic behind the self-targeting of workfare, where the forgone asset is an outside job (Besley and Coate, 1992; Ravallion, 1991), and of opting out of public schools (Besley and Coate, 1991); I show that this extends to in-kind transfer programs and document it directly for a housing transfer.

Second, this paper contributes to the literature studying housing for the urban poor. Programs like providing titles (Field, 2005, 2007; Galiani and Schargrodsky, 2010), construction materials (Cattaneo et al., 2009; Galiani et al., 2017), or improved public services (Michaels et al., 2021; McIntosh et al., 2018) lead to improvements in specific components of housing. Despite such measures the urban poor persistently live in slums over many years. Marx et al. (2013) suggest that slum dwellers might need a "big push" to grow out of poverty and live in improved houses. Interventions that move households elsewhere disrupt their economic networks and see low take up (Picarelli, 2019; Franklin, 2020; Barnhardt et al., 2017). Public reconstruction programs are plagued with delays and many slum dwellers do not receive the promised house (Philip, 2026). In

contrast, I show that beneficiaries of this *in situ* program experience large improvements in house quality — consistent with the big push hypothesis and without the disruption costs of relocation. The improved house in turn serves as an adaptation against extreme heat and heavy rainfall.

Third, this paper makes a methodological contribution to the welfare analysis of public programs. A large literature in public finance recovers welfare from reduced-form sufficient statistics rather than fully specified structural models (Chetty, 2009). I show that when participation is voluntary and preferences have an extreme-value shock, the take-up rate and the dispersion of preferences are sufficient statistics for the expected welfare a program delivers, whatever the functional form of utility. This is the program-evaluation counterpart of the ACR characterization of the gains from trade (Arkolakis et al., 2012), with the non-take-up rate in place of the domestic expenditure share and the dispersion of preferences in place of the trade elasticity. The approach complements the marginal-value-of-public-funds literature, which values marginal policy changes through their welfare and fiscal effects (Hendren and Sprung-Keyser, 2020; Finkelstein and Hendren, 2020); by contrast, my framework values a discrete, non-marginal offer, including the surplus of inframarginal participants net of their participation costs. Because the dispersion parameter only scales welfare, the welfare-maximizing benefit level can be recovered from take-up rates observed or elicited at different benefit levels, making the method applicable to other programs with incomplete take-up, such as workfare.

Finally, to my knowledge, this is the first well-identified evaluation of PMAY, India's largest housing program, which accounts for 1–2% of the national budget and targets 20 million poor urban households. The evaluation has three features that make it informative for policy. First, it studies the program at scale rather than a pilot (Muralidharan and Niehaus, 2017). Second, the program was designed and implemented entirely by the government with no involvement from researchers, so the estimates capture its effects as actually delivered; effects of researcher- or NGO-implemented programs are often larger than those of the same programs run by governments (Bold et al., 2018; Vivalt, 2020; List, 2022). Third, it spans 11 towns across the state rather than a single site selected for evaluation (Allcott, 2015).

2 Context and Intervention

2.1 Context

This project is based in Odisha, India. Odisha is one of the poorer states of India with a per capita GSDP of 2,200 USD in 2025-26. The state had a population of 42 million in 2011 of which 16.7% lived in urban areas. There are 114 urban local bodies in the state and in this project I focus on the 109 municipalities and notified areas in the state which accounted for about 61% of the urban population in 2011.[4]

Slums are a common feature of most cities and towns in India. As of 2011, 17% of all urban households lived in slums across the country. In Odisha, 23% of urban households lived in slums (Census of India, 2011). The Government of Odisha defines a slum as a compact settlement of at least twenty households with a collection of poorly built tenements, mostly of temporary nature, crowded together usually with inadequate sanitary and drinking water facilities in unhygienic conditions. The state government has identified nearly 1500 slums across these cities and towns. About 160,000 households or about 500,000 people live in these slums.

Slums develop organically as people migrate from rural to urban areas. Increasing structural transformation has led people move to towns and cities in search of better employment opportunities. Such people set up a temporary shelter at vacant lands in the city. They usually start with a house made of tarpaulin sheets and progressively build a more formal structure using bamboo sticks, mud, metal and asbestos sheets, and bricks (Rains and Krishna, 2020).

The slum households do not usually have any formal land documents. They are squatters on the land. However, over time they build different forms of de facto and de jure land rights. The de facto land rights come from their neighbours and other slum dwellers recognizing their claim to the land. This even allows the households to sell their land in informal land markets and transfer such de facto rights. Households may also hold different documents that tie them to their land. Households usually start with public utilities connections like electricity and water. They may also get some informal documents from the local governments. It is very difficult for slum dwellers to get full land rights for their land though (Krishna et al., 2020).

Because their housing is informal, slum dwellers face two threats. First, they face a threat of eviction by the government and other agencies. Slum dwellers are evicted from their lands whenever

[4]The study does not include the 5 municipal corporations where the program implementation had not begun until 2022.

new developments are planned in those lands. They are usually left homeless by such actions and need to restart at a new location. Second, slum dwellers live in very poor and unhygienic conditions. In the context I study, about 8% households have a concrete roof while the rest have asbestos or polythene roofs. 56% have walls made of bricks while the rest have walls made of mud or bamboo. 30% households have a toilet in their house and 60% have access to any toilet. The poor living conditions are a health hazard to the slum dwellers. Many share toilets with other households. They are susceptible to water-borne and communicable diseases. Their roofs are not resistant to heavy rain and their houses flood during rains. They also face a threat of pests during the rains. Children may also not be able to pursue education if they do not have a conducive environment and resources to study at home. Marx et al. (2013) suggest that these limits on human capital accumulation together with inertia in physical capital investments may create a poverty trap for these households.

Policy responses to slums have usually taken the form of different housing programs, improvements in public infrastructure, and land titling programs. In the next subsection I discuss the intervention in this study.

2.2 Pradhan Mantri Awas Yojana (PMAY)

Pradhan Mantri Awas Yojana (translates to Prime Minister's Housing Scheme; PMAY) is an Indian central government program to provide formal houses to the poor. The program succeeds earlier housing programs, including the Indira Awas Yojana (1985) for rural households and the Rajiv Awas Yojana (2013) for urban slum dwellers. The program was launched in 2015 with the aim of ensuring all urban households have a formal house. Under the urban component of the program, the government propose to provide houses to over 20 million families and over 12 million houses have been sanctioned as of 2026. The program is partly funded by the state governments and implemented by the state governments through the municipal administrations. The rural and urban components of the program combined accounts for 1.38% of the union government's budget for the year 2026-27.[5]

PMAY includes multiple strategies to provide housing including subsidized housing credit, government-led construction of housing shelters, and beneficiary-led construction. I focus on the third component which also accounts for over 60% of the houses sanctioned under the program

[5]In the years 2016-2024 the government has allocated Rs 3.89 trillion or about 49 billion USD for the combined implementation in rural and urban areas across India.

as of 2026. Under the beneficiary-led construction scheme, beneficiaries are given Rs 150,000 to construct a house on their own land. Some state governments, including the Government of Odisha, top up the grant with an additional Rs 50,000 to make the total grant worth Rs 200,000 (∼USD 2,500). This is about twice the median annual household in the slum sample in this study.

To be eligible to receive the benefits, the households must meet three criteria: (i) They need to belong to the Economically Weaker Sections category or have a total household annual income of less than Rs 300,000. Nearly all slum dwellers meet this requirement.[6] (ii) The households should have a land title or formal tenure security for the land where they propose to construct the house. (iii) The family should not own a house or land elsewhere in the country.

The households wishing to benefit from the scheme need to submit a set of documents to their local municipal authorities. The authorities then submit the request on a central portal. Some households receive a work order which confirms that they will receive money under the program. The household can then start construction. They need to break down any existing dwelling structure to lay a foundation. They must then raise a plinth and build a concrete roof. The households receive the money in 4 installments; after completion of the foundation, walls, roof, and the entire house. At the completion of each stage of construction the municipal authorities verify the status of construction. They upload geo-tagged photographs of the site to a portal. The households then receive money directly in their bank accounts.

The government grant covers a part of the cost incurred by the households. The grant amount of Rs 200,000 may be sufficient to construct a basic house in a small town. The households also incur costs in temporary stay arrangements during the construction phase. While some households stay with their families or neighbors, others find a house for rent temporarily or construct a small temporary shelter near their land. The households also need to incur expenses up front and receive the grant money later on.

2.3 Odisha Government's JAGA Mission

To avail the house construction grant from PMAY households need to have a land title or tenure documents for their land. Most slum dwellers are squatters and have no formal land titles. Many of them have lived in the same land for a very long time and gain de facto land rights. They even manage to get electricity and water connections. But they do not have any documents that guarantee their tenure on that land. The Government of Odisha introduced a program to give land

[6] According to the data collected by the government, all households report an annual income of Rs 250,000 or less.

tenure to these slum dwellers. There is variation in which slum dwellers have received these land rights and in turn variation in their eligibility for the housing grant. I use this variation to evaluate the causal impact of the housing grant. In this section I describe the program and in the next section how this yields identification for the causal impact.

The Government of Odisha launched JAGA Mission in 2017 with the aim of transforming all slums in the state into livable habitats with all civic infrastructure at par with the rest of the city. The government embarked on a four-pronged approach to achieve this. In the first phase, the government planned to provide tenure security to the slum dwellers through land rights certificates. In the second phase the government intended to provide basic civic amenities like electricity, water, sewage, roads, and public shelters in all slums. In the third phase, the slum dwellers were to be encouraged to build a formal house on their lands taking advantage of the house construction grants from PMAY. In the fourth phase, the slums were to be made an active part of planning development activities in their neighborhood through slum dweller associations.

The government passed a legislation in 2017 authorizing them to provide land rights certificates to all slum dwellers. A household with a land rights certificate (LRC) cannot be evicted from that land. This gives the households formal tenure security. However, unlike land titles, the land rights certificates are non-transferable. The households cannot sell, sublease or gift the land to anyone else. They can only bequeath the land to their heir. To implement the program, the government first imaged all slums using drones. They mapped each parcel of land with a family through a household survey. The land parcels data was also linked with the revenue department data to identify the official owners of the land. All households living on land owned by the municipal government or abandoned land immediately received the land rights certificates. The certificates were issued in the name of the woman of the house whenever possible.

The formal land owners are varied. Some of these lands were vacant. Some were owned by the local municipal government or state government departments like municipal administration or public works department. In these cases the government could immediately give the slum dwellers the land rights certificate. I will henceforth refer to these lands as *tenable* lands in this paper. The other lands were owned by private individuals, temple trusts, erstwhile royal families, and central government departments like railways and forests. In these cases, the government can try to get the lands transferred from the other departments to the municipal government and then give the households the land rights certificates. In the case of privately owned lands the government cannot give the households land rights certificates for the same land and plans to move the households to

a different government owned land. I will henceforth refer to these lands as *untenable* lands in this paper. As of January 2023 the government was still in the process of identifying and acquiring new lands to move these households. Less than 5 slums had been shifted to new lands at that time.

3 Empirical Strategy

3.1 Identification strategy

To estimate the causal impact of receiving the house construction grant I use the variation in the implementation of the tenure security program. I compare the households living on *tenable* lands with the households living on *untenable* lands within the same slum. As described in Section 2, the households living in tenable lands became eligible to receive tenure security immediately from the municipal government. Only households who have received tenure security documents were eligible for the house construction grants.

For this identification strategy to be valid, the two groups of households need to be comparable and tenability exogenous to the outcomes. When migrants first arrive in the city, they occupy vacant lands near existing slums or other vacant lands to set up tents. Over time they settle down there and build a permanent structure. They do not know who owns the vacant land at the time of settling down. A slum is usually less than a few hundred meters wide. Within this small area, it is difficult to ascertain which lands are tenable and which are untenable without access to detailed revenue department data. There is unlikely to be any selection by slum dwellers into tenable or untenable lands. Figure 3 shows the interspersed distribution of tenable and untenable households within one slum.

The slum dwellers living in the same slum are likely to face similar economic opportunities and environmental conditions. There is little variation across households within the same slum in access to employment opportunities and public services. Their relative location within the city is also the same. The slum dwellers in the same slum would also face a similar level of eviction threat. Evictions are usually implemented at the slum level rather than the individual household level. The slum dwellers living in tenable and untenable lands should have had a similar trajectory over time as well.

I test for similarity between households living in tenable and untenable lands in observables before the intervention. I estimate the coefficient β in the following equation

$$y_{is} = \delta \; tenable_{is} + \Gamma_s + \epsilon_{is} \tag{1}$$

where y_{is} is the outcome for household i in slum s before the implementation of these interventions; $tenable_{is}$ is a binary variable that is 1 if the household lives on a tenable land and 0 if they live on an untenable land; Γ_s are slum fixed effects. The balance tables are described in Section 3.4.

Tenability affects households through two channels: it provides tenure documents, and through them, eligibility for the grant. Interpreting the IV estimates as the effect of the grant therefore requires that tenure documents alone do not affect the outcomes. Section 4.5 shows that households receiving tenure documents but not the grant experienced no detectable improvement in house quality.

3.2 Data sources

3.2.1 Instrument data

The Government of Odisha conducted a drone based imaging of all slums across Odisha prior to the implementation of the tenure security intervention in 2017. They identified the areas occupied by the slum houses. They then mapped the households with the extent of the land occupied by them in a household survey. The drone images were also matched with the revenue records of the cities to determine the formal owner of the land. I use this data on the formal owner to classify households as living on tenable or untenable lands.[7] In the case of many households, their plot lies at the intersection of multiple revenue plots with different owners. In these cases the household is classified as tenable if more than half the house lies on tenable lands and as untenable otherwise. In the full sample 54% of households live on tenable lands.

3.2.2 Pre-intervention data

The government conducted a household survey of all slum dwellers along with the drone imaging in 2017 prior to the distribution of tenure documents. In this survey the households were asked about the total annual household income, characteristics of their house, demographic details of all

[7] I classify as tenable lands where the Record of Rights is abandoned land (*anabadi* or *patita*), municipality (*pourasanstha*), reserved for public works (*sarbasadharana* or *rakshita*), animal husbandry (*gochar*), water department (*jalachar*), or health department. The untenable lands are those where the Record of Rights names private individuals, lands of erstwhile royal estates, railway departments, and forest department.

household members, and whether the family belongs to the economically weaker sections category. This data also includes the area of the land that the household occupies. This data is used to verify exogeneity of the instrument, as controls in all regressions, and to measure heterogeneity in incidence by baseline outcomes.

3.2.3 Outcomes data

I conducted two rounds of surveys of these slum dwellers to measure outcomes 6 years and 9 years after the start of the program to distribute tenure documents. The surveys were conducted in February-April 2023, and February-April 2026 respectively. Both rounds were administered in-person by enumerators. The sample included 5,917 households in round 1 and 6,001 households in round 2 from 201 slums across 11 cities in Odisha.

11 urban local bodies out of 109 where the program intervention was underway were chosen for the survey. These include Balangir, Baripada, Bhawanipatna, Brajrajnagar, Dhenkanal, Jeypur, Jharsuguda, Keonjhargarh, Koraput, Malkanagiri, and Rairangpur. These towns had a population of 50,000–150,000. They are spread across the state as seen in Figure 2 and cover different parts of the state.

There are 483 slums in all in these cities from which 201 were included in the survey sample. Appendix B explains how slums were selected. From each sample slum 30 households were selected at random for the survey with replacements for households who could not be reached. In the second round of data collection we managed to track 93% of the respondents from round 1 and there is no differential missingness by tenability. The missing households were replaced with other houses.

The surveys included measures of program receipt, house quality, economic indicators like income, employment, consumption, savings, and loans, health and education outcomes, and measures of assets in the house. During the survey the households were also asked their willingness to take up the program if it was offered at different levels of grant amounts ranging from Rs 50,000 to Rs 500,000 in increasing order.

3.2.4 Program implementation data

The tenure security program is implemented by the local government bodies. They maintain a record of all households who have received the tenure documents. I obtained this record as of Aug 2022 for all the cities in my sample. In the survey, households were asked what tenure documents they had and whether they had received the grant.

3.3 Estimation strategy

To analyze the incidence of the program and estimate the progressivity of the incidence, I estimate the coefficients in this specification:

$$Y_i = \gamma \text{IHS(income)}_i + \epsilon_i \tag{2}$$

where Y_i is an indicator for program take up, IHS(income)_i is the inverse hyperbolic sine transformation of income prior to the introduction of the program.

I estimate the causal impact of receiving the tenure documents and the house construction grant on outcome measures. I estimate the ITT impact parameter as the coefficient in the specification

$$Y_{is} = \beta \; tenable_{is} + \Gamma_s + \mu \; x_{is}^{BL} + \epsilon \tag{3}$$

where Y_{is} is the outcome measure, $tenable_{is}$ is an indicator for household living on a tenable land, and Γ_s are slum fixed effects. x^{BL} is a set of covariates from before the intervention.

I also present the impact of receiving the PMAY grant on all outcomes using the tenability of the land as an instrument. These estimates would be the causal estimate if the tenability of the land affects only the receipt of the grant and not through any other channel. This requires that just receiving the tenure documents alone did not have an effect on any of the outcomes directly. In Section 4.5 I present evidence for the direct effect of receiving the tenure documents being small.

3.4 Validity of the Research Design

3.4.1 Instrument exogeneity

In Table 2 I present the balance in covariates prior to the intervention for households surveyed in the second round. For most variables the difference in covariate values across tenable and untenable households is not statistically significant. I can reject differences greater than 3-5 percentage points for most[8] house quality measures and greater than 10% of the mean for measures like household income. Differences greater than 3 years for years since when the household has lived in that slum are also rejected. Appendix Table D2 presents the corresponding balance table for respondents in the first round of the survey. The baseline characteristics are added as covariates in the regression.

[8]The share of houses with brick walls differs (3.8pp, p-value: 0.03); this is one of 16 covariates, and baseline wall material is included as a control in all regressions.

In Appendix Table D1 we see the balance in covariates across tenable and untenable households among the set of all slum households in these 11 cities. The difference in covariate values across tenable and untenable households is not statistically significant for 11 of the 12 variables. The larger sample size allows me to reject even smaller differences. Differences larger than 1.5pp for the house quality measures and differences larger than 0.05 sd for all other variables are rejected. This strongly suggests that the instrument used is exogenous.

3.4.2 Attrition

Attrition from the survey sample is one of the concerns in follow-up surveys conducted six and nine years after the initial survey. The only identifiers from the census conducted by the government available are the names of the household members. This further reduces the probability of finding the houses. Attrition from the survey itself only reduces precision but differential attrition across tenable and untenable households within the same slum can lead to biased estimates.

During the first round of the survey, enumerators were handed a list of 30 households drawn from the government census to survey. They were also given a list of up to 30 alternative households from which they could pick households to replace those who could not be surveyed in the first 30. Of the primary list of 6,030 households, surveys could be completed with 4,827 households (80%). We see in Table C1 that the response rate was not significantly different across untenable and tenable households (difference of 2.6pp, p-value of 0.09). The remaining households could not be surveyed because of two primary reasons: the household refused or was unavailable (8.8%), or the household had migrated (6.3%).[9]

In the second round of the survey, 93% of the respondents of the first round were surveyed again. The difference in attrition across untenable and tenable households is nearly 0 (p-value: 0.97; Table C2).

[9]The households living on tenable lands were slightly more likely to have migrated than those living on untenable lands (difference of 1.9pp, p-value of 0.04). This is driven by permanent migration (and not temporary migration). However the point estimate is small and the balance table shows that the untenable and tenable households in the survey sample are balanced across most characteristics.

4 Results

4.1 Program Incidence

Program incidence is progressive even within the survey sample: households with lower pre-intervention income are more likely to have received and taken up the grant. In the first column in Table 1, a one-unit increase in baseline IHS income (inverse hyperbolic sine transformation of income) lowers the probability of take-up by 2.3 percentage points (significant at the 1% level), against a mean take-up rate of 21%. In the sample, the gap between the 95th percentile household and the 5th percentile household is 2 units of IHS baseline income and the drop in program receipt is 4.6 pp, about a fifth of the mean. This incidence reflects both households' own decisions to participate and the government's eligibility and selection criteria.

To isolate household preferences, I examine three stated-preference measures (Table 1, columns 2–7). The first measure is whether the household applied, among households that were asked by a local official to apply (column 2). This application probability falls by a similar 2.2pp (p-value: 0.07) for every unit of IHS income. Second, when asked whether they would take up the program at the current benefit level, the share saying yes declines by 5.0 percentage points per unit of IHS income (column 4), from a mean of 51%. Third, the benefit level at which households would take up the program rises with income by about Rs 22,000 per unit of IHS income (column 6)—richer households demand a larger grant to participate. All three point to poorer households self-selecting into the program.

This self-selection is driven partly by the quality of the household's house before the intervention. To receive the grant a household must demolish its existing house and rebuild, so those with worse houses face a lower opportunity cost and are more likely to participate. Consistent with this, the income gradient in each preference measure attenuates once I control for pre-intervention house quality (columns 3, 5, and 7 in Table 1): the gradient in intended take-up falls from 5.0 to 3.6 percentage points per unit of IHS income, and the application gradient from 2.2 to 1.6. The baseline house quality measures account for about a quarter of the observed gradient across all measures.[10]

[10]The preference gradients nonetheless remain progressive even conditional on pre-intervention house quality. This may reflect additional filters in the government's eligibility criteria, or that the observed house-quality measures are incomplete and income is correlated with unobserved house quality.

4.2 Treatment effects on program receipt

Table 3 presents the first stage of the instrument. Tenable households are 14.3 percentage points more likely than untenable households in the same slum to report having received tenure documents by 2026 (Panel B; column 1; untenable mean 8%).[11] However, according to government records (Panel A; column 2) tenable households were 41 percentage points more likely to have received the tenure documents as of Aug 2022. The difference between the two measures may be because the government did not deliver the documents or the households did not remember receiving them at the time of the survey.

Tenable households are 12.0 percentage points more likely to report having received the PMAY housing grant (Panel B; column 2; untenable mean 9%) and have received on average Rs 18,458 more in grant money than untenable households in the same slum (column 3; untenable mean Rs 11,712). They are also 5.9 percentage points more likely to have completed construction of the new house (column 4; untenable mean 4%). All four first-stage effects are significant at the 0.1% level.

We see that the receipt of tenure document is steady from 2023 to 2026 (Panel A to Panel B). However, households continued to receive the grant until 2026. 72% of the recipients reported having received the grant before 2024, 14% in 2024, 12% in 2025, and 2% in 2026 before the survey. Hence the short-run outcomes measured in 2023, when tenable households were 9.8 percentage points more likely to have received the grant but only 2.5 percentage points more likely to have completed construction, largely capture households in the middle of construction. By 2026, about half of the additional recipients had completed their house.

The t-statistic for the coefficient of tenable in the regression with grant receipt as outcome is 9.4 (corresponding F-statistic is 89), which allays concerns about weak instruments. Because the first stage on receiving the grant is about 12 percentage points, the ITT effects on downstream outcomes are roughly an eighth of the effect of receiving the grant; the corresponding IV estimates are reported in Appendix F.

4.3 Treatment effects on house quality

The primary outcome is the endline quality of the house. A house quality index is defined as the first principal component of many characteristics of the house including material of the roof, walls, floor, toilets, safety features, and number of rooms and windows. In Table 4, the ITT effect

[11] Untenable households may receive tenure documents if the department gets the land transferred from the other departments.

of tenability on the house-quality index is 0.088σ, significant at the 5% level (Panel B; column 1). The effect in 2023 is smaller (0.051σ) and not statistically significant, consistent with many recipients still being in the middle of construction, often living in temporary accommodation. The impact on house quality is considerable given the small first stage effect on receiving the grant (12 percentage points). The impact on the house quality index persists even when we define the index differently (Table E3). The treatment effect of receiving the grant itself, measured using IV regressions, is 0.75σ (Table F1).[12]

Columns 2–6 report ITT estimates on individual components of the house. The improvement is concentrated in a few dimensions of house quality: the probability of having a concrete roof rises by 9.6 percentage points (column 2, significant at 1%), and there is a smaller, statistically insignificant effect on having brick walls, covered floors, and toilets. There is no improvement in the safety features of the house like having concealed electric wires or safer doors. When the grant is disbursed, the second installment is paid after they construct a brick wall, and the third installment after they lay a concrete roof. Hence, the monitored components of the house improve while we do not see significant improvements in the other components at the time of the survey. It is possible that the households would add features to their house slowly over time. The impacts on all features of the house are presented in Tables E1 and E2.

The improvement in concrete roofs is considerable. The ITT estimate constitutes a 50% increase in probability of having a concrete roof. The IV estimate of the impact of receiving the grant itself is 80 percentage points. Having a concrete roof in itself is an aspiration for many households and is viewed as a status symbol. This is reflected in the treatment effect on self-reported satisfaction with their house (0.16 points on the 1–5 satisfaction scale, significant at 1%; Table 5; column 1).

A concrete roof is also an important adaptation against extreme weather. We find an improvement in protection against heavy rainfall and intense heat. They are 4 percentage points more likely to report no leak from the roof when it rains heavily (Table 5; column 2). The protection from heat also improves among households living on tenable lands. The difference between the temperature outside their house and inside their house is a measure of how much protection the house offers against extreme heat[13]. When the outside temperature exceeds 32℃, the gap between outside and inside temperature is 0.22℃larger for tenable households (column 5, significant at 10%), consistent with better insulation from the concrete roof. The corresponding IV estimate is 1.9℃ reduction

[12]The IV estimates assume that receiving the tenure documents itself does not lead to an improvement in house quality. Section 4.5 provides evidence for this.

[13]This measure was pre-specified.

in inside temperature relative to the outside temperature from receiving the grant. The effect is nearly zero on cooler days when the outside temperature is less than 32℃. The effects are likely driven by the concrete roof. Figure 5 shows the descriptive relationship between temperature inside and outside by whether they have a concrete roof or not. This reduction in temperature inside could have considerable health benefits including a reduction in heat-induced mortality (Bressler et al., 2026).

4.4 Treatment effects on other downstream measures

I find no detectable effects on pre-specified downstream outcome measures of health, incomes, or assets by 2026. These are intent-to-treat estimates with a 12 percentage point first stage, so they are informative about modest effects of tenability but cannot rule out moderate effects of receiving the grant itself. Table 6 reports effects on measures of health and education. Among health outcomes there is no effect on the incidence of fever, and the only statistically significant estimate in the table is a 3.8-percentage-point increase in diarrhoea among tenable households. This is an implausibly large effect and is possibly due to noise in the data. Education outcomes—the share of children under 18 in school, in private institutions, and in English-medium instruction—show no significant effects, nor do the fertility outcomes of under-five mortality and the number of pregnancies.

The program leaves the household's broader economic circumstances unchanged in the medium run (Table 7): there is no significant effect on income, the presence of a business, consumption, or the value of household assets. The one economically sizable estimate is on outstanding debt—tenable households carry about Rs 10,189 more in outstanding loans, roughly 28% above the untenable mean, though this is imprecisely estimated (not significant at conventional levels). The point estimate rises from 17% of the untenable mean in 2023 to 28% in 2026, consistent with more households borrowing as they constructed. To the extent households borrow to finance construction beyond the grant, the resulting debt-servicing burden could offset any income or consumption gains from the improved house, which would help explain the absence of effects on those margins. Households might accrue income or consumption benefits after they repay these loans.

4.5 Impact of tenure security alone

The treatment effects estimated above are the combined effect of receiving tenure security documents and the house construction grant. In this section I explore how much of the effect on house quality comes from tenure security itself, and how much from the construction grant it unlocks.

I decompose the two and find that the grant does nearly all the work: households receiving only tenure security improve their house by at most a tenth compared to households receiving the grant as well.

Tenure documents can directly impact households in a few ways. First, households may perceive that they own the land and can sell the land and the house at any point. Second, households may be able to mortgage their tenure documents to raise money. Third, the tenure documents may reduce the households' perceived threat of eviction from the slum and encourage them to invest more in their house. In Table 8 we see that the first two channels may be supported by fewer than 1% of the households. However, there is a significant 5.5pp reduction in the perceived threat of eviction (from a base of 34%).

I present a difference-in-differences estimate to separate the effect of receiving tenure security only from receiving the grant as well. Restricting to households who responded in both rounds of the survey, and to house quality components measured in the baseline and in the two endline surveys (roof, walls, and toilet), I create a panel of house quality measures over three periods. The households are grouped by the programs they received.

In Figure 4 we see that there was an improvement over time among households who did not receive either program. Households who received tenure documents but not the grant follow almost exactly the same path as households who received neither. The households who received PMAY grant before 2023 start at a lower house quality level in 2017 but see a significant increase in house quality relative to the previous two groups in both time periods. Households who received PMAY grant after 2023 and before 2026 see parallel trends in house quality until 2023 before a significant improvement in 2026. This suggests that the house quality improvements were primarily due to the grant and not just the tenure documents alone.

To quantify the effect and account for slum fixed effects, I estimate the following specification:

$$\Delta H_{is} = \delta_T T_i + \delta_{TG} TG_i + \delta_G G_i + \theta_{q_i} + \Gamma_s + \epsilon_{is} \tag{4}$$

The households are classified into four groups based on what they had received by 2026: neither a land-rights certificate nor the grant (N), a certificate only (T), both a certificate and the grant (TG), and the grant without a recorded certificate (G).[14] The outcome ΔH_{is} is the change in house quality of household i in slum s between the 2017 baseline and 2026, measured using the house

[14]These households were recorded as not having received tenure documents as of 2022 but may have received subsequently. It is not possible to receive the grant without having a tenure document.

features recorded in both surveys. θ_{q_i} is the baseline house quality quintile fixed effect. Γ_s is the slum fixed effect. The model is estimated using all households responding in the second round.

We might be worried about households selecting into taking up the grant based on their current house quality. To allay this concern, the coefficients are also calculated separately for each quintile by interacting the quintile fixed effect with each group.

Table 9 presents coefficients estimated using OLS. In the pooled regression we see a 0.5σ improvement in house quality over the nine years among households who received neither program. Households who received the tenure documents alone improved house quality by a further 0.08σ (p-value: 0.11). Households who received PMAY as well improved house quality by 1.15σ (p-value < 0.001) relative to households who received neither. We can conclude from this that the receipt of the tenure documents alone can only account for about 7% of the observed improvement in house quality. This result holds up among households in all 5 baseline house quality quintiles: the point estimate for receiving tenure alone is less than a fifth of receiving the grant as well and statistically significant only for the second quintile.

The improvements in house quality are also driven by roof and walls which are the metrics targeted by the grant. There is a small or no improvement on all other metrics. This further suggests that the improvement is driven by the receipt of the grant and not just the tenure documents.

5 Welfare and optimal grant size

Consider a government with a fixed budget for this program. Should it give a smaller grant to more recipients or a larger grant to fewer recipients? This is a frequently encountered question in the design of welfare benefits (Hanna and Olken, 2018). Diminishing marginal utility suggests spreading the transfer across many households and giving each household a small transfer. However, the literature on big push interventions finds that a large one-time transfer helps move people beyond a poverty-trap threshold and leads to lasting benefits (Balboni et al., 2022; Banerjee et al., 2015). The need for large transfers is especially true with housing which is a lumpy investment. A formal house requires a large upfront payment while the benefits accrue over many years (Marx et al., 2013).

In this program, the grant size also changes the composition of people who apply for it. As the grant size increases, households who have a better house and hence face a larger participation cost also take up the grant. These households gain less overall because much of the grant's value is used

to offset the participation cost. This selection effect supports a smaller grant even when utility is linear in money.

Identifying the optimal grant size under these conditions requires a way to estimate the welfare gains. However, such housing programs provide benefits across a wide range of outcomes and over a very long time horizon. While it is difficult to capture the full impact on all welfare outcomes, the households' take-up decision aggregates all these effects. In this section, I provide a framework to estimate welfare using only the household's preference to take up the program. The key idea is to translate the willingness to participate in the program into a measure of average welfare benefits from the program as in the Roy model. I then use this framework and a take-up preference elicitation to evaluate the optimal grant size.

5.1 Setup

Consider a household with income I_i and house quality h_i at baseline. The household has a choice to take up PMAY benefits with benefit level b. If they take up the program, they will receive utility $u_{yi}(b) + \lambda_i$ and if they do not take up the program they will receive utility u_{ni}. The terms u_{ni} and u_{yi} are assumed to be a function of the household's baseline characteristics alone while the term λ_i is an idiosyncratic preference shock that an individual household receives. We can think of the utility values as the maximized utility values received after maximizing over all other choices. This allows the setup to abstract away from other household decisions.

The key assumption is on the distribution of λ_i.

Assumption 1. *The idiosyncratic preference term is distributed according to a logistic distribution:* $\lambda_i \sim$ *logistic*$[0, \beta]$

The difference of two independent random variables with a Gumbel distribution has a logistic distribution. Hence, this assumption is equivalent to assuming that the utilities u_{yi} and u_{ni} each carry an additive idiosyncratic shock drawn independently from a Gumbel (type-I extreme value) distribution. The Gumbel distribution assumption is natural here because u_{yi} and u_{ni} are maximized value functions. Each aggregates many underlying household choices like how much to spend on housing versus other consumption, the type of construction, arrangements during construction, etc. Each of these choices is subject to its own idiosyncratic shock. A random variable which takes the maximum of many independent shocks converges to a Gumbel distribution. Analogous extreme value distributional assumptions underpin discrete choice models (McFadden, 1974), quantitative

trade models (Eaton and Kortum, 2002), and migration models (Bryan and Morten, 2019). In Appendix H, I show that the qualitative conclusions from this section are not sensitive to this assumption and hold if we instead assume the shocks are distributed normally.

The parameter β is proportional to the standard deviation of the distribution and is a measure of the heterogeneity in preferences. A higher value of β means that the households with the same baseline income and house quality differ more in the utility they receive from the housing grant.

Households take up the grant if their utility from taking up is greater than not taking up or $u_{yi} + \lambda_i > u_{ni}$.[15] Hence the welfare gain that a household receives due to the option to take up the program is given by

$$W_i = \max(u_{yi} + \lambda_i - u_{ni}, 0) \tag{5}$$

This is the welfare gained in the intent to treat sense, which is 0 for non-takers and the actual benefit for those who take up the program. This assumes that any household who wants the program gets it. We will address this assumption later in this section.

5.2 Sufficient statistics for welfare

Given baseline characteristics of the household, the terms u_{yi} and u_{ni} are fully determined and the choice of taking up the program only depends on λ_i. The probability (over the idiosyncratic shock) that the household takes up is

$$p_i = \Pr(\lambda_i > u_{ni} - u_{yi})$$

Under assumption 1, we can simplify the probability expression as in the proposition below.

Proposition 1. *The probability p_i that a household with given baseline characteristics takes up the program is given by*

$$p_i = \frac{e^{u_{yi}/\beta}}{e^{u_{yi}/\beta} + e^{u_{ni}/\beta}} = \frac{1}{1 + e^{(u_{ni} - u_{yi})/\beta}} \tag{6}$$

The proof, in Appendix G, follows from the properties of the logistic distribution. We can now use this probability to simplify the expected welfare gain. From equation 6 we have

$$u_{yi} - u_{ni} = \beta \log\left(\frac{p_i}{1 - p_i}\right) \tag{7}$$

[15] This presumes households can assess the returns to participation. This is reasonable in this setting, where the program has been running for several years and households have observed neighbors move through construction and into completed houses.

Using this, we can express the expected welfare gain to any household in terms of their take up probability. The expectation is again over only the idiosyncratic preference shock.

Proposition 2. *Under assumption 1, the expected welfare transferred to a household i is given by*

$$E[W_i] = \beta \log\left(\frac{1}{1-p_i}\right) \tag{8}$$

Proof. See Appendix G. ■

The proposition states that the average welfare transferred is only a function of the probability of participating in the program and β. Thus (p_i, β) is a sufficient statistic for the expected welfare gain due to the program for a household. This formula is also analogous to the ACR formula for welfare gained from trade. The take up probability is analogous to the share of own trade and the spread parameter β is analogous to the inverse of the trade elasticity.

The sufficient statistic result is intuitive: if the welfare gain from a program is high then every household would want to take up the program while if the welfare gain is moderate or close to zero on average then only a smaller share of households would be willing to take up. Assuming the distribution of the idiosyncratic shocks allows us to translate the proportion to an exact welfare value. If we assume a different distribution, the exact relationship will change while the qualitative result would still hold.

Further, this framework is relevant in situations where a significant share of the population does not want to take up the program (or p_i is much less than 1 for many households). As $p_i \to 1$, the expected welfare diverges and its sensitivity to the take up rate increases without bound. However, in a program like PMAY where a modest share of the population reports not wanting to take up the benefits of the program, this is a reasonable theoretical framework.

This allows us to also define the aggregate welfare gain among the entire population as:

$$E[W] = \sum_{i=1}^{n} E[W_i] = \sum_{i=1}^{n} \beta \log\left(\frac{1}{1-p_i}\right) \tag{9}$$

Now consider a scenario where the government has a fixed budget B and the grant per beneficiary is b. The government can give the program to at most B/b households. However, the number of households wanting to take up the grant may be more. In this case I assume that the government randomly chooses recipients from among the applicants.[16] Thus the probability that a household

[16] Governments try to award grants to all who demand them. But in practice only a share of the population

applying for the grant gets it is $\min(1, \frac{B}{bn_a})$ where n_a is the number of households applying for the grant. Since the lottery probability does not depend on the expected welfare gain, the total welfare gain is given by:

$$E[W] = \min\left(1, \frac{B}{bn_a}\right) \sum_{i=1}^{n} E[W_i] = \min\left(1, \frac{B}{bn_a}\right) \sum_{i=1}^{n} \beta \log\left(\frac{1}{1-p_i}\right) \tag{10}$$

5.3 Optimal benefit level

Suppose the government has a fixed budget B for the program. It is important to identify the grant size b that optimizes welfare. As the grant size increases, three changes happen simultaneously: (a) each existing recipient receives a larger transfer and hence higher welfare gain; (b) households with a higher participation cost (or a higher value of u_{ni}) would also select into taking up the program. This will reduce the overall welfare gain since the gain for these recipients is small; (c) the total number of beneficiaries decreases. Hence it is theoretically ambiguous how increasing grant size would affect overall welfare gain and depends on the distribution of u_{ni} in the population.

We can empirically estimate the welfare gain using the above framework and a take-up preference elicitation conducted during the endline survey. During the survey, each household was presented with a series of hypothetical grant sizes from Rs 50,000 to Rs 500,000 and asked at each whether they would take up the program. This traces out their stated take-up as a function of the benefit level. All other elements of the program remained the same and only the grant size varied. Responses were internally consistent: all households switched at a single threshold, taking up at all higher grant sizes and none lower; and threshold benefit levels moved predictably with baseline characteristics, rising with house quality and income, indicating good comprehension of the question.

To use the above framework, we need a probability of take up conditional on baseline characteristics. I group households by the baseline income quintile and their baseline house quality quintile. This creates 25 groups of about 200-300 households each. For each household I estimate the take up probability as the share of households in that group who take up the program. I then use expression in Equation 10 to estimate the total welfare gain to the entire population.

This estimator is valid under the assumptions of the framework. By construction, u_{yi} and u_{ni} are functions of baseline income and house quality alone, so households in the same cell share the same deterministic utilities and differ in their take-up decisions only through the idiosyncratic shock

receive such benefits. Lotteries are a common mechanism used in allotting benefits in housing programs. Due to data limitations governments are not able to target specific characteristics.

λ_i. Because λ_i is drawn independently of observable characteristics, the share of households in a cell who take up is a sample mean of independent draws with mean p_i, and is therefore an unbiased estimator of the take-up probability. The parameter β multiplies every household's expected welfare and scales the welfare curve without changing its shape. Hence the welfare-maximizing grant size does not depend on β.

Figure 6 presents the welfare gain at different grant sizes. The government budget is held constant at half the amount disbursed to the households in the survey until 2026. When the grant size is Rs 50,000, the budget is not fully spent. However, when the grant size increases from Rs 100,000 to Rs 300,000, the welfare gain decreases. This is due to a lot of households with a high participation cost (high u_{ni}) selecting into participating in the program. Subsequently, the increase in benefit per household dominates the selection effects and welfare increases slightly. Importantly, this analysis suggests that if the government has limited budget for the program, reducing the grant size can lead to more welfare gain due to an increase in the number of beneficiaries and better screening out of households with a higher baseline house quality.

In Appendix H I show that these conclusions remain qualitatively the same even if we assumed that the idiosyncratic preference shocks were distributed normally instead of logistically.

6 Discussion

Improving housing conditions in slums is a central policy goal across the developing world, and one that will only grow as rural–urban migration continues. The recurring concern with housing programs is that they either fail to reach the poorest or deliver little for those they do reach, particularly when participation requires households to co-invest and to demolish their existing dwelling. This paper studies a large *in situ* program: India's PMAY, at 1–2% of the federal budget, and finds that it does neither: its incidence is progressive and its medium-run impact on house quality is large.

The program is progressive despite its ordeals. The probability of receiving the grant falls by 4.6 percentage points from the 5th to the 95th percentile of the baseline income distribution, against a mean take-up of 21%. This progressivity is driven by poorer households self-selecting into the program. The key participation cost is the value of the existing dwelling a household must demolish to rebuild: richer households have invested more in their current house and so face a higher opportunity cost of participating. Consistent with this, about a quarter of the income

gradient in the stated take-up operates through baseline house quality rather than income directly. This mechanism of an income-correlated participation cost created by forfeiting an existing asset generalizes beyond housing to any program where participants give up an outside option, such as workfare (the forgone job) or school-voucher programs (the forgone school).

By the medium run, nine years after tenure documents were distributed and after most construction is complete, the program delivers a large improvement in house quality: a LATE of 0.75σ among grant recipients. The short-run estimates from the first survey round were considerably smaller, because households living through the construction phase temporarily occupy worse accommodation. The effect is concentrated almost entirely on the government-verified margin, the probability of a concrete roof, with little detectable effect on unverified dimensions such as toilets, windows, or wiring. Since each installment is conditioned on verified milestones, the program improves precisely what it monitors. These physical gains translate into tangible flow benefits: recipients report higher satisfaction with their house, are better protected from rain, and live in cooler interiors on hot days.

Despite a transfer worth roughly twice annual household income, these gains do not extend to broader economic and human-capital outcomes in the medium run. I find no significant effects on income, consumption, assets, health, education, or fertility. Part of the explanation may be financial: recipients carry about 28% more outstanding debt (though imprecisely estimated), likely borrowed to finance construction beyond the grant, and the associated repayment burden could offset income or consumption gains until it is retired. More fundamentally, housing is a large, lumpy investment with a substantial participation cost, the value of the demolished dwelling, which reduces the net worth of the transfer to the household. This is common to infrastructure programs, where incremental improvement is often infeasible and gains require a large upfront outlay.

The improvement is driven by the construction grant rather than by tenure documents on their own. Households that received tenure documents but not the grant improved their houses no more than households that received neither, and the gains are concentrated in the concrete roof that the grant finances. For these households, the binding constraint on housing investment appears to be the capital required to build a formal house rather than the partial tenure security these documents provide, consistent with a "big push" view (Marx et al., 2013).

Finally, the welfare framework points to a concrete design lever. Because the take-up rate is a sufficient statistic for the average welfare the program transfers, I can use households' elicited willingness to take up at different grant sizes to compare policies under a fixed budget. The

comparison favors a smaller grant spread over more households: this both covers more beneficiaries and screens toward those with the most to gain whereas a larger grant draws in households with better houses and small welfare gains. For a program of PMAY's scale, even a modest reduction in the grant size could raise the aggregate welfare delivered.

Several questions remain open. Whether the absence of income and consumption effects is temporary, and will reverse once construction loans are repaid, or reflects a genuinely limited pass-through from housing to livelihoods is a question for a longer-run follow-up. As the program scales, general-equilibrium effects on local construction costs and land values may become important. The study covers small and medium-sized towns in a lower-income state; in large cities, where land and construction costs are higher and the grant covers a smaller share of the cost of a formal house, both take-up and impacts may differ. Finally, the JAGA Mission in Odisha drew on external agencies for components such as land mapping and community mobilization, in addition to the state's own capacity. Replicating these results elsewhere may depend on similar support for handing out tenure guarantees, which makes households eligible for the grant.

7 Tables

Table 1: PMAY incidence by baseline income

	Grant received	Applied | selected		Intend to take up		Threshold benefit (Rs)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Baseline income (IHS)	-0.023**	-0.022	-0.016	-0.050***	-0.036***	22168***	16659***
	(0.007)	(0.012)	(0.012)	(0.009)	(0.009)	(3767)	(3802)
BL house-quality controls	No	No	Yes	No	Yes	No	Yes
Mean of outcome	0.210	0.818	0.818	0.511	0.511	269238	269238
N	6001	2369	2369	6001	6001	6001	6001

This table presents how the incidence of PMAY varies with baseline income. The dependent variable in the first column is whether the household had received the program when surveyed in the second round in 2026; in the next two columns is whether they applied for the program conditional on being encouraged to do so by a local government official; in the fourth and fifth columns is whether they stated that they would like to take up the program if offered to them; and the last two columns is the lowest benefit level at which they would be willing to take up the program. The independent variable is the inverse hyperbolic sine of their reported income prior to the introduction of the program. The independent variable is 10.3 at the 5th percentile and 12.3 in the 95th percentile. Six measures about the roofs, walls, and toilets of the house prior to the program are added as controls in columns 3, 5, and 7. Standard errors are reported in parentheses. $^{*}\ p < 0.05$, $^{**}\ p < 0.01$, $^{***}\ p < 0.001$

Table 2: Balance table

	Untenable		Tenable				
	n	mean	n	mean	Diff	Confidence interval	p-value
Plot chars							
Occupied area (sqft)	1847	492.45	4154	506.57	20.265	-3.736 - 44.266	0.10
Area per person (sqft)	1847	186.16	4154	188.47	-0.758	-12.961 - 11.444	0.90
Income							
Annual income (Rs.)	1847	58813.47	4154	58494.21	-48.113	-3,605.878 - 3,509.651	0.98
Income per capita (Rs)	1847	20733.23	4154	21615.31	-604.246	-2,193.908 - 985.416	0.46
House quality							
Concrete roof	1847	0.06	4154	0.09	0.002	-0.018 - 0.022	0.85
Concrete/tiled roof	1847	0.30	4154	0.29	-0.009	-0.040 - 0.021	0.55
Brick wall	1847	0.52	4154	0.58	0.038	0.003 - 0.073	0.03*
Toilet in house	1847	0.32	4154	0.32	0.009	-0.023 - 0.041	0.57
Direct access to road	1847	0.76	4154	0.67	0.009	-0.019 - 0.037	0.54
Demographics							
HH size	1847	3.31	4154	3.33	0.095	-0.012 - 0.201	0.08
Number of children	1847	0.93	4154	0.95	0.080	-0.004 - 0.165	0.06
Average household age	1847	33.26	4154	33.12	-1.073	-2.161 - 0.015	0.05
Proportion female	1847	0.51	4154	0.50	-0.015	-0.034 - 0.003	0.10
Education							
Highest adult edu	1794	6.85	4035	7.33	0.224	-0.132 - 0.580	0.22
Mean adult edu	1797	5.08	4048	5.39	0.178	-0.117 - 0.473	0.24
Migration							
Years in slum	1759	59.57	3953	55.56	-0.286	-2.656 - 2.083	0.81

Notes. The sample includes all households surveyed in the endline. The first 4 columns present the mean and number of observations for individuals in tenable and untenable lands respectively. The last three columns present the coefficient of regressing the outcome on the tenability of the land with slum fixed effects, the 95% confidence interval, and the p-value. We can reject differences between the two groups larger than the confidence intervals.
$^{*}\, p < 0.05$, $^{**}\, p < 0.01$, $^{***}\, p < 0.001$

Table 3: First stage: Program receipt by tenability

Panel A: Short-run impacts

	Received tenure document		PMAY		
	(1)	(2)	(3)	(4)	(5)
	Self-report	Govt record	Received	Money received	Construction complete
Tenable	0.134***	0.414***	0.0983***	11187.0***	0.0246***
	(0.0144)	(0.0151)	(0.0111)	(1679.2)	(0.00642)
Untenable mean	0.09	0.08	0.04	5480.13	0.01
Slum FE	Yes	Yes	Yes	Yes	Yes
N	5826	5878	5917	5905	5917

Panel B: Medium-run impacts

	Received tenure document	PMAY		
	(1)	(2)	(3)	(4)
	Self-report	Received	Money received	Construction complete
Tenable	0.143***	0.120***	18458.0***	0.0585***
	(0.0140)	(0.0127)	(2197.3)	(0.00930)
Untenable mean	0.08	0.09	11711.96	0.04
Slum FE	Yes	Yes	Yes	Yes
N	6001	6001	5979	6001

Notes. This table presents the ITT impact of living on tenable lands on program receipt. The coefficients estimated are those in equation 3 estimated using OLS. The dependent variables are the household's report of whether they have received tenure documents, the government's records of whether they have received tenure documents (as of Aug 2022), whether the household has received any benefits under the house construction grant, the amount of money received per household (unconditional on program receipt), and whether they have completed PMAY house construction. All regressions include slum fixed effects and pre-intervention covariates. Panel A presents results from the first round in 2023 and Panel B presents results from the second round in 2026. $^{*}\ p < 0.05$, $^{**}\ p < 0.01$, $^{***}\ p < 0.001$

Table 4: ITT effects on house quality

Panel A: Short-run effects

	House quality index	House quality components				
	(1)	(2)	(3)	(4)	(5)	(6)
		Roof	Wall	Floor	Toilet	Concealed wires
Tenable	0.0514	0.0583***	0.0172	-0.00427	0.0213	0.0166
	(0.0321)	(0.0140)	(0.0169)	(0.0157)	(0.0155)	(0.0173)
Untenable mean	0.00	0.14	0.66	0.71	0.57	0.39
Slum FE	Yes	Yes	Yes	Yes	Yes	Yes
N	5910	5917	5917	5917	5917	5917

Panel B: Medium-run effects

	House quality index	House quality components				
	(1)	(2)	(3)	(4)	(5)	(6)
		Roof	Wall	Floor	Toilet	Concealed wires
Tenable	0.0881*	0.0962***	0.0209	0.0274	0.00355	-0.000828
	(0.0352)	(0.0148)	(0.0157)	(0.0153)	(0.0151)	(0.0179)
Untenable mean	-0.00	0.18	0.76	0.76	0.66	0.44
Slum FE	Yes	Yes	Yes	Yes	Yes	Yes
N	5744	6001	6001	6001	6001	6001

This table presents the ITT estimates of living on tenable lands on the house quality index and some components of the index. The dependent variables are the house quality index defined as the principal component of all individual measures and indicators for whether the house has a concrete roof, brick wall, covered floor, toilet in the house, and concealed wires which are safer. The estimates are coefficients from specification 3. All regressions include slum fixed effects and baseline controls. Panel A presents results from the first round in 2023 and Panel B presents results from the second round in 2026. $^{*}\ p < 0.05$, $^{**}\ p < 0.01$, $^{***}\ p < 0.001$

Table 5: ITT effects on benefits of a better house

Panel A: Short-run effects

	Satisfaction with house	Rainfall	
	(1)	(2) No roof leak	(3) No waterlogging
Tenable	-0.00591	0.0234	0.0308*
	(0.0597)	(0.0180)	(0.0141)
Untenable mean	3.91	0.27	0.85
Slum FE	Yes	Yes	Yes
N	5905	5917	5917

Panel B: Medium-run effects

	Satisfaction with house	Rainfall		Outside − inside temp	
	(1)	(2) No roof leak	(3) No waterlogging	(4) All days	(5) Hot days
Tenable	0.156**	0.0399*	-0.00528	0.0768	0.223†
	(0.0572)	(0.0174)	(0.0155)	(0.0545)	(0.118)
Untenable mean	3.84	0.23	0.80	1.53	1.59
Slum FE	Yes	Yes	Yes	Yes	Yes
N	5987	5829	5829	5903	1851

This table presents ITT estimates of living on tenable lands on measures of benefits from their house. The dependent variables are self-reported satisfaction with their house, whether their roof did *not* leak and whether there was *no* waterlogging during the last rainy season, and the difference in the temperature outside and the temperature inside (℃). The last column restricts to households surveyed on days when the outside temperature was above 32℃. Panel A presents results from the first round in 2023 and Panel B presents results from the second round in 2026. † $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 6: ITT effects on health, education, and fertility outcomes

Panel A: Short-run effects

	Health		Education			Fertility	
	Fever	Diarrhoea	In school	Private	English	U5MR	Pregnancies
Tenable	0.00894	0.00204	0.00728	-0.00248	0.00177	-0.0462	0.0188
	(0.0124)	(0.0127)	(0.0189)	(0.0158)	(0.0189)	(0.0294)	(0.0297)
Untenable mean	0.24	0.08	0.66	0.17	0.19	0.22	0.38
Slum FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	5917	3731	3947	3947	3125	1630	5917

Panel B: Medium-run effects

	Health		Education			Fertility	
	Fever	Diarrhoea	In school	Private	English	U5MR	Pregnancies
Tenable	0.00489	0.0378*	-0.0152	-0.0111	-0.00853	-0.0141	0.0349
	(0.0136)	(0.0148)	(0.0189)	(0.0168)	(0.0191)	(0.0304)	(0.0281)
Untenable mean	0.31	0.11	0.74	0.21	0.20	0.18	0.31
Slum FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	6000	3599	3762	3762	3128	1358	6001

This table presents ITT estimates of living on tenable lands on health, education, and fertility outcomes. The dependent variables are whether each adult member had a fever in the last month, whether each child had diarrhea in the last month, share of children below 18 years of age in school, share of children in private schools, share of children in english-medium schools, under-5 mortality rate, and the number of pregnancies in the last 5 years. The sample in columns 2-5 includes only households with children below age 18 and in column 6 includes households with at least one pregnancy in the last 5 years. Panel A presents results from the first round in 2023 and Panel B presents results from the second round in 2026. $^{*}\, p < 0.05$, $^{**}\, p < 0.01$, $^{***}\, p < 0.001$.

Table 7: ITT effects on economic outcomes

Panel A: Short-run effects (round 1)

	Income	Any business	Consumption	Asset value	Outstanding loans
Tenable	587.4	0.00866	634.3	-391.6	6646.9
	(602.0)	(0.0182)	(335.8)	(849.6)	(4106.1)
Untenable mean	12091.81	0.32	8978.60	36686.81	39221.02
Slum FE	Yes	Yes	Yes	Yes	Yes
N	5897	5910	5917	5903	5917

Panel B: Medium-run effects (round 2)

	Income	Any business	Consumption	Asset value	Outstanding loans
Tenable	-300.1	-0.00467	116.3	-1170.5	10188.7
	(811.1)	(0.0177)	(418.4)	(1188.0)	(8002.7)
Untenable mean	14167.80	0.29	11261.30	54325.93	36209.14
Slum FE	Yes	Yes	Yes	Yes	Yes
N	6001	6000	5919	5993	6001

This table presents ITT estimates of living on economic outcomes. The dependent variables are total household monthly income, whether any household member runs a business, total household monthly consumption, total value of assets in the house, and total value of outstanding loans. Panel A presents results from the first round in 2023 and Panel B presents results from the second round in 2026. $^{*}\, p < 0.05$, $^{**}\, p < 0.01$, $^{***}\, p < 0.001$.

Table 8: Impact of tenure documents alone

Panel A: Short-run effects

	Received tenure doc	Have some tenure doc	Threat of eviction	Tried to mortgage	Able to mortgage	Able to sell land
	(1)	(2)	(3)	(4)	(5)	(6)
Tenable	0.136***	0.120***	-0.0547**	0.00147	0.00283	-0.00658
	(0.0144)	(0.0168)	(0.0174)	(0.00270)	(0.00184)	(0.00528)
Untenable mean	0.09	0.17	0.34	0.00	0.00	0.02
Tenable mean	0.29	0.40	0.29	0.01	0.00	0.01
Slum FE	Yes	Yes	Yes	Yes	Yes	Yes
N	5824	5824	5850	5894	5894	5894

Panel B: Medium-run effects

	Received tenure doc	Have some tenure doc	Threat of eviction	Tried to mortgage	Able to mortgage
	(1)	(2)	(3)	(4)	(5)
Tenable	0.143***	0.141***	-0.0559**	0.00211	-0.00114
	(0.0139)	(0.0169)	(0.0185)	(0.00206)	(0.00158)
Untenable mean	0.08	0.19	0.53	0.00	0.00
Tenable mean	0.31	0.42	0.43	0.01	0.00
Slum FE	Yes	Yes	Yes	Yes	Yes
N	6001	5853	5895	5861	5861

Notes. This table presents the ITT effects on variables related to receipt of tenure security documents. The coefficients are those in specification 3. The dependent variables are the household's report of whether they have received tenure documents, the household's report of having any tenure security document (those received under this program as well as from any other authority), their perceived threat of eviction from that land, whether they have tried to mortgage their tenure documents, their perceived ability to mortgage their tenure documents, and (Panel A only) their perceived ability to sell their land. All regressions include slum fixed effects and pre-intervention covariates. Panel A presents results from the first round in 2023 and Panel B from the second round in 2026.

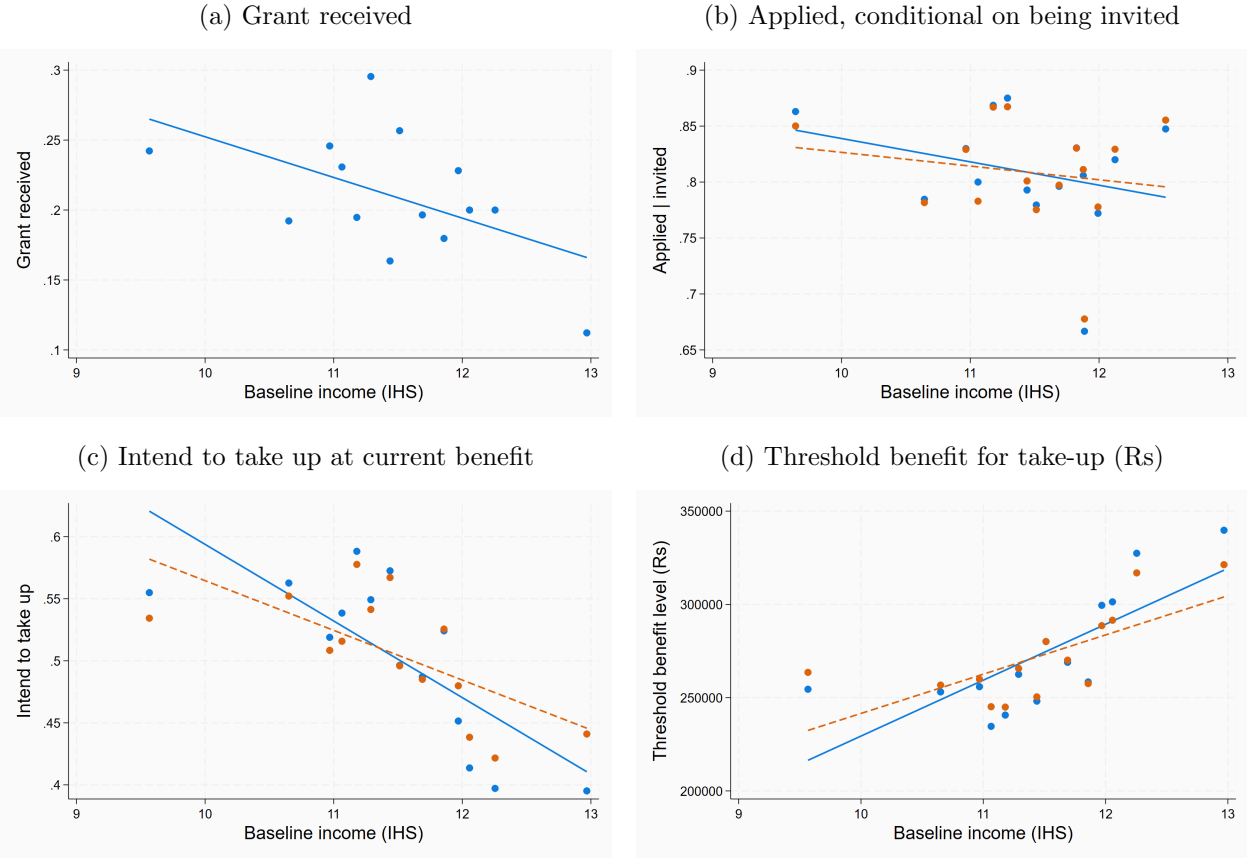
Table 9: Separating the effect of tenure from the grant: change in house quality, 2017–2026

	Effect on change in house quality				N		
	Tenure alone	Grant given tenure	Grant, no LRC	Change, neither	LRC only	LRC + grant	Grant, no LRC
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Quintile 1 (worst)	0.098	1.267***	1.113***	1.344	391	406	67
	(0.085)	(0.117)	(0.256)				
Quintile 2	0.192**	0.932***	0.865***	0.571	541	360	61
	(0.071)	(0.086)	(0.142)				
Quintile 3	-0.040	1.377***	0.884***	0.292	156	76	24
	(0.095)	(0.157)	(0.258)				
Quintile 4	0.068	0.923***	0.774***	0.283	312	148	21
	(0.094)	(0.140)	(0.202)				
Quintile 5 (best)	-0.011	0.831***	0.856***	-0.222	213	103	46
	(0.086)	(0.143)	(0.203)				
Pooled (quintile FE)	0.079	1.067***	0.918***	0.498	1613	1093	219
	(0.049)	(0.061)	(0.113)				

Notes. The outcome is the change in house quality from 2017 to 2026, measured using features recorded in both surveys. Rows for each quintile are from a single regression interacting group indicators with baseline house-quality quintiles (equation 4 with group-by-quintile interactions); the pooled row is from equation 4. Groups are defined by the government land-rights certificate (LRC) record and self-reported grant receipt; households with neither are the omitted group. Column 1 is the coefficient on LRC only. Column 2 is the difference between the LRC + grant and LRC-only coefficients. Column 3 is the coefficient on grant without an LRC record. Column 4 is the mean change among households with neither. Columns 5–7 report group sizes. All regressions include slum fixed effects. Standard errors clustered by slum in parentheses. $^{*}\,p < 0.05$, $^{**}\,p < 0.01$, $^{***}\,p < 0.001$.

8 Figures

Figure 1: PMAY incidence and stated take-up by baseline income



Notes. Binned scatter plots of each outcome against the inverse hyperbolic sine of pre-intervention household income. The 5th and 95th percentiles of baseline income are 10.3 and 12.3. The outcome variables are (a) whether the household had received the program when surveyed in the second round in 2026, (b) whether the household applied for the program conditional on being encouraged to do so by a local government official, (c) whether they stated that they would like to take up the program if offered to them, and (d) the lowest benefit level at which they would be willing to take up the program. Solid lines and blue markers show the raw relationship. Dashed lines and orange markers control for six baseline measures of the roof, walls, and toilet. Fitted slopes are the coefficients in Table 1. Panel (b) is restricted to households invited to apply by a local official.

Figure 2: Map of Odisha with cities in the sample

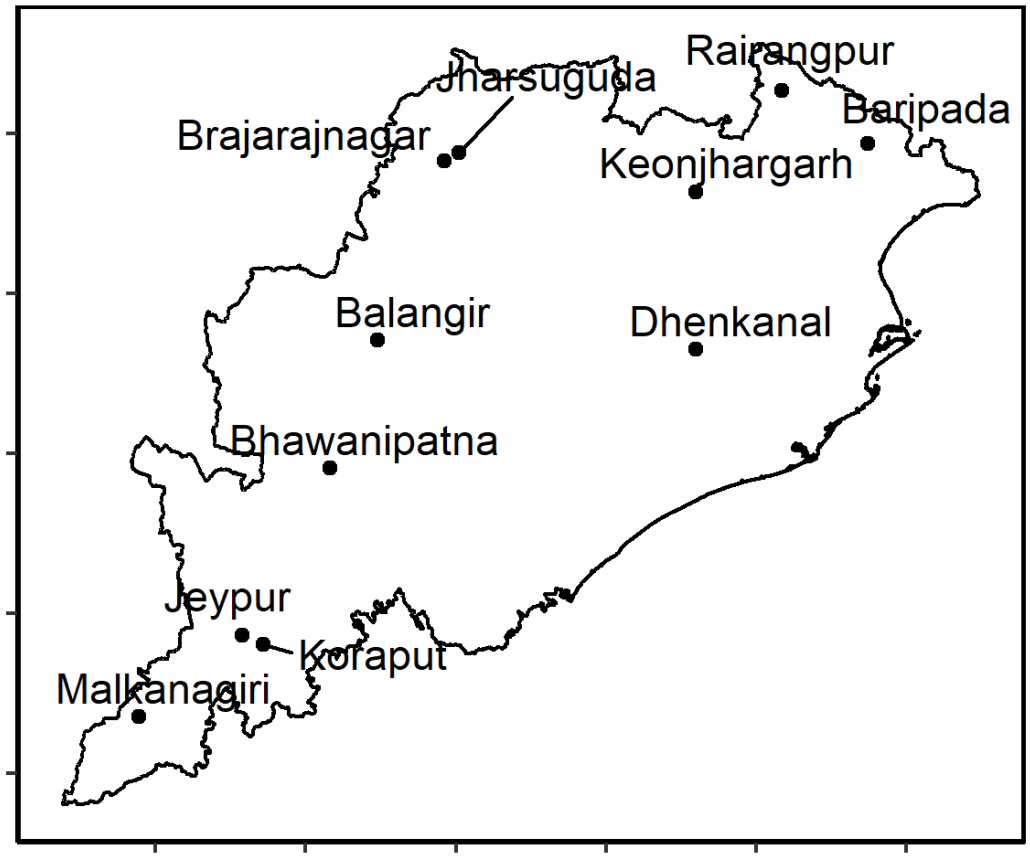


Figure 3: Distribution of tenable and untenable households in one slum

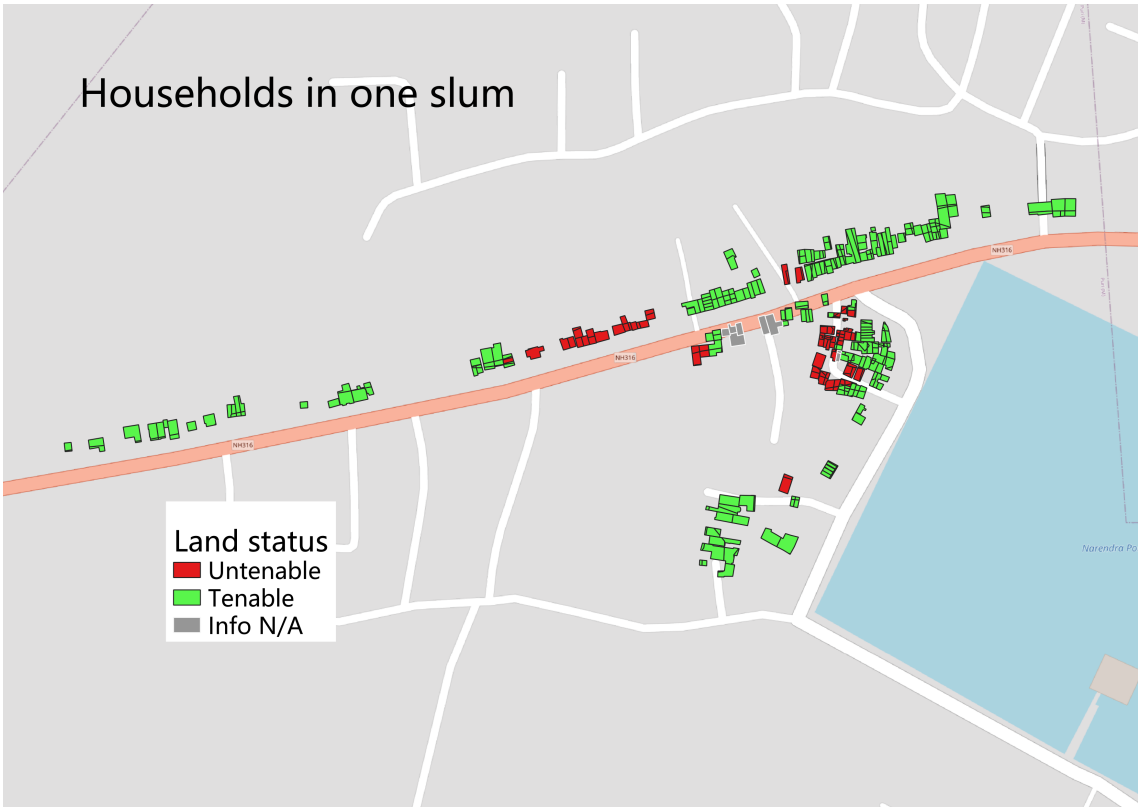
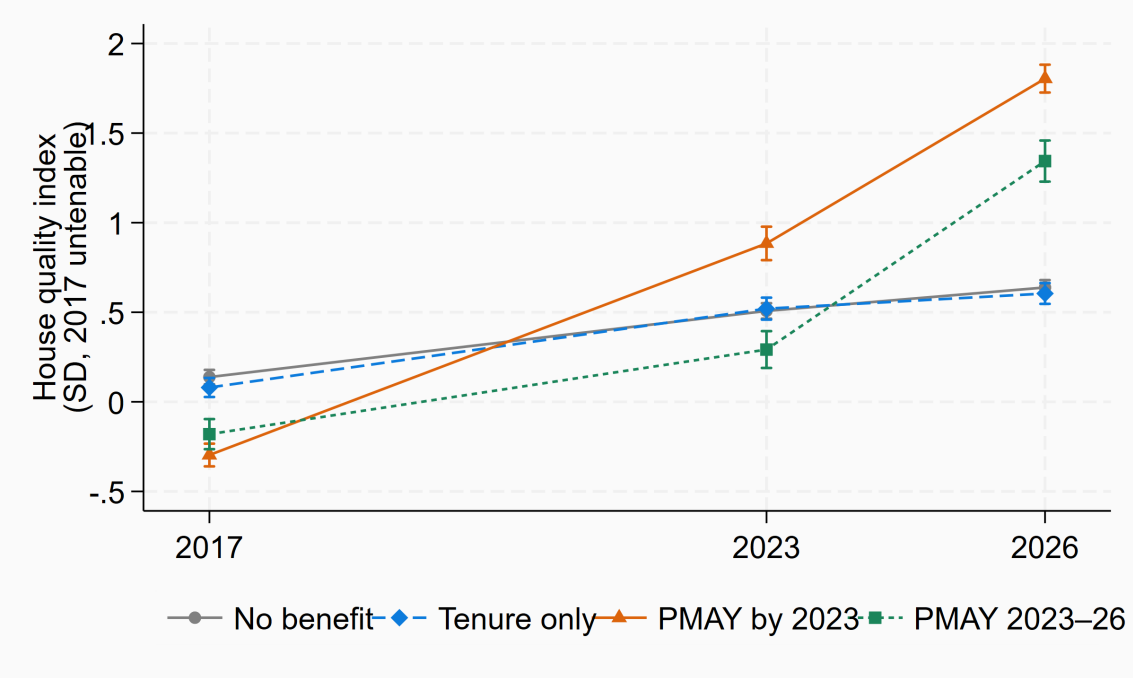
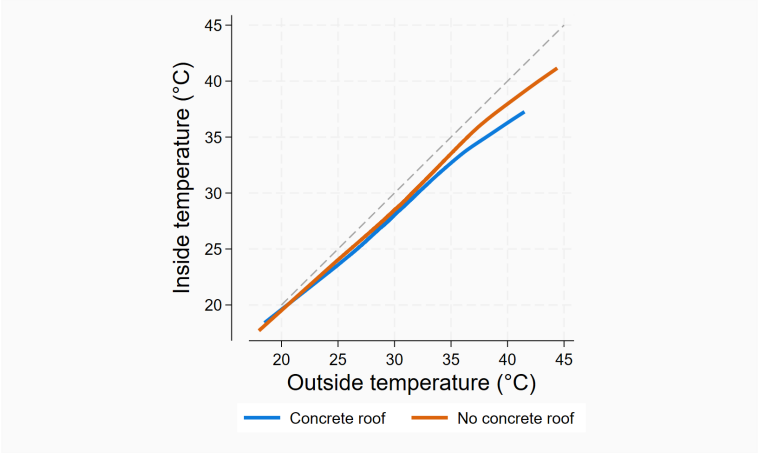


Figure 4: House quality over time by tenure and grant receipt



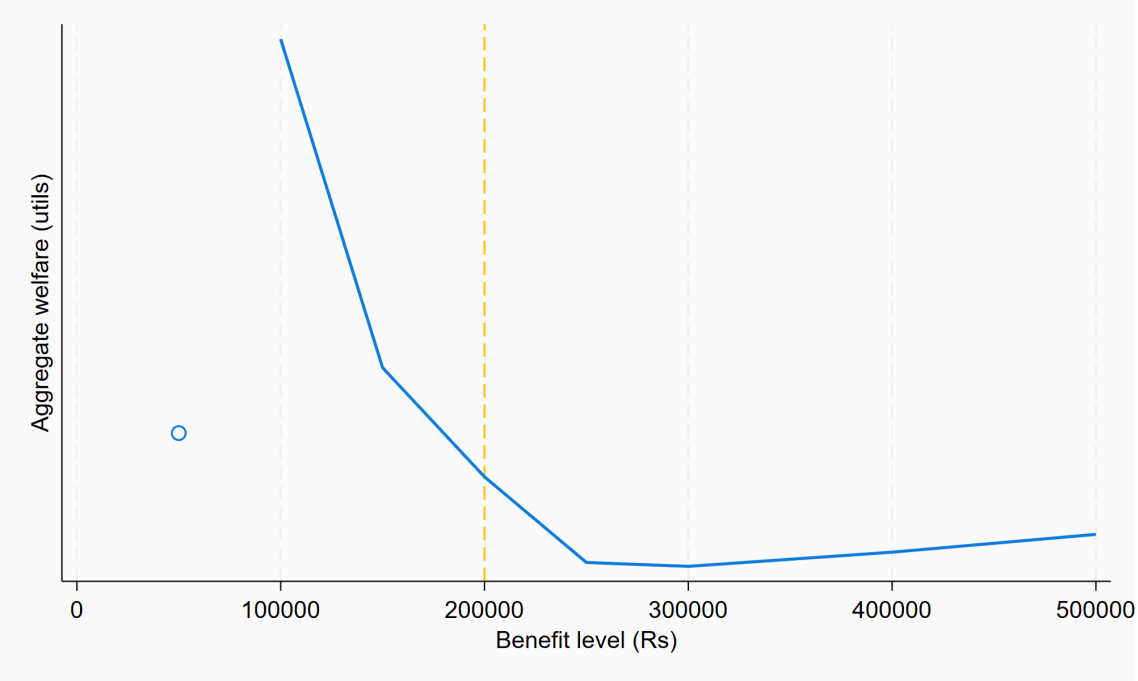
Notes. This figure plots mean house quality in the 2017 baseline census, the 2023 survey and the 2026 survey for the balanced panel of households surveyed in both rounds. House quality is a common index built from features measured in all three years (roof, wall and toilet), using baseline principal-component weights, and standardized relative to untenable households in 2017. Households are grouped by program receipt: no tenure document and no grant (No benefit); a land-rights certificate but no grant by 2026 (Tenure only); the grant received by 2023 (PMAY by 2023); and the grant received between 2023 and 2026 (PMAY 2023–26). Bars show 95% confidence intervals. Means are unadjusted; Table 9 reports the corresponding estimates with slum fixed effects within baseline house-quality quintiles.

Figure 5: Inside and outside temperature by roof type



Notes. This figure plots locally weighted (lowess) regressions of the temperature inside the house on the temperature outside, measured by enumerators at the time of the second survey round in 2026, separately for houses with and without a concrete roof. The dashed line is the 45-degree line, along which inside and outside temperatures are equal. Houses are compared descriptively and differ in characteristics other than the roof; the causal effect of the grant on the inside–outside temperature difference is reported in Tables 5 and F2. Few survey days had outside temperatures above 40℃.

Figure 6: Welfare gain at different grant sizes



Notes. This figure presents estimated aggregate welfare transferred at different benefit levels when the idiosyncratic preference shock is distributed $Logistic(0, \beta)$. Take-up probabilities are cell means of stated take-up from the 2026 survey, with cells defined by baseline income quintile and baseline house-quality quintile. The budget is fixed at half the current level and grants are rationed by lottery when demand exceeds the number of grants. The hollow marker at Rs 50,000 is a corner case in which take-up is low and part of the budget goes unspent. The dashed line marks the current benefit of Rs 200,000.

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Appendix A: Selection into the program by quintiles

The main-paper incidence analysis (Table 1) regresses each selection outcome on continuous IHS baseline income. Table A1 reruns the same regressions with baseline income-quintile dummies (quintile 1 being the base category).

Table A1: Selection into the program by baseline income quintile

	Grant received	Applied | selected		Intend to take up		Threshold benefit (Rs)	
Income quintile 2	0.0103	0.0359	0.0384	0.0255	0.0284	-975.6	-1891.3
	(0.0167)	(0.0251)	(0.0251)	(0.0204)	(0.0203)	(8583.2)	(8539.3)
Income quintile 3	-0.0247	-0.0373	-0.0324	-0.0434*	-0.0331	12755.2	8975.6
	(0.0151)	(0.0227)	(0.0226)	(0.0185)	(0.0185)	(7801.6)	(7777.0)
Income quintile 4	-0.0489**	-0.0128	-0.00460	-0.0187	-0.00549	3804.4	-1626.6
	(0.0185)	(0.0278)	(0.0277)	(0.0227)	(0.0226)	(9553.4)	(9510.5)
Income quintile 5	-0.0410*	-0.0181	-0.00591	-0.131***	-0.0934***	63675.8***	48998.5***
	(0.0175)	(0.0270)	(0.0272)	(0.0215)	(0.0218)	(9034.4)	(9192.9)
BL house-quality controls	No	No	Yes	No	Yes	No	Yes
N	6001	2369	2369	6001	6001	6001	6001

Notes. This table presents how the incidence of PMAY varies across quintiles of baseline income, using the second survey round in 2026. Each column is a separate OLS regression of the outcome on indicators for baseline income quintiles 2 to 5; the first quintile is the omitted category. The dependent variable in column 1 is whether the household has received the program. In columns 2 and 3 it is whether the household applied for the program, conditional on being encouraged to apply by a local government official. In columns 4 and 5 it is whether the household states that it would take up the program at the current benefit of Rs 200,000. In columns 6 and 7 it is the lowest benefit level at which the household would take up the program (Rs); households that would not take up the program at any benefit level offered are assigned Rs 1,000,000. Columns 3, 5 and 7 control for six measures of the roof, walls and toilet of the house prior to the program. Regressions do not include slum fixed effects. Standard errors in parentheses. $^{*}\ p < 0.05$, $^{**}\ p < 0.01$, $^{***}\ p < 0.001$.

Appendix B: Slum sample selection

There are 109 urban local bodies in Odisha where the program had been implemented as of 2022. Among these, data was available for 25 cities. 11 cities were chosen to be in the sample based on ease of surveying, the number of slums in the city, and the spread across the state.

There are 483 slums in these cities. Reliable data was available for 333 slums from these cities and this forms the sampling frame for this survey. The sample selection was done to meet two objectives: estimate the treatment effects within a slum net of any general equilibrium effects and measure general equilibrium effects by comparing slums with fewer and more tenable households. However, there were imbalances in pre-intervention covariates across slums with fewer tenable households and slums with more tenable households.

Slums with data available for fewer than 30 households are dropped. I define a slum level propensity score to predict the share of tenable households in a slum using the baseline covariates. This was done by regressing the share of tenable households in a slum against slum level aggregates of pre-intervention covariates about house quality, household income, and demographics and city fixed effects among slums in all cities. These coefficients were used to define the propensity score as the predicted share of tenable households. Within each city slums were sorted by propensity scores to create strata of 4 slums with similar propensities.

To be able to compare households within the same slum we need to over-sample slums which have an equal share of tenable and untenable households. To achieve this, I define a slum as being balanced if it has between 10% and 85% of households living on tenable lands. In each stratum there are at most 4 slums. All strata which have 3 or 4 balanced slums are included in the sample. 97 slums are chosen this way. Among the remaining strata, 31 are chosen at random adding another 104 slums to the sample.

Appendix C: Attrition and migration

C.1 Attrition from baseline to round 1

This section presents the rate at which households could be tracked in the first round of the survey. The sample frame included all houses who were surveyed by the government during the census in 2017-18. A primary sample of 30 households were chosen in each sample. The enumerators tried to survey as many of these households as possible. For households who were unreachable, replacement households were picked from outside the list of 30 households.

Table C1 presents the share of tenable and untenable households among the primary sample of 30 households by survey status. We see in the first row that there is no statistically significant difference in the probability of being surveyed by the tenable status. The two types of households have a similar likelihood of not being surveyed for most of the reasons. The likelihood of having migrated is similar in magnitude for tenable and untenable households but the difference is statistically significant when we add slum fixed effects. The baseline level of migration is very low and the point estimate is not economically meaningful.

Table C1: Attrition

	Untenable	Tenable	Diff	p-value
Survey complete	.83	0.79	-0.026	0.09
All dead	.015	0.01	-0.005	0.23
Merged with another	.015	0.02	-0.006	0.20
Refused	.073	0.09	0.016	0.14
House not found	.0094	0.03	0.002	0.77
Migrated	.061	0.06	0.019	0.04*

Notes. Each row in the table corresponds to a survey status (survey completed or reason for non-completion). The first two columns are the proportion of tenable and untenable households with that survey status. The third column is the difference between tenable and untenable proportions estimated with a slum fixed effect. The fourth column gives us the p-value of the difference being different from 0.

C.2 Attrition from round 1 to round 2

This section presents the rate at which households could be tracked in the second round of the survey. Among the households with whom surveys were completed in the first round of the survey, 93% could be tracked and surveyed again in the second round. There was no differential missingness by tenable status. Table C2 presents the share of tenable and untenable households by survey status in the second round.

Table C2: Attrition: round 1 to round 2

	Untenable	Tenable	Diff	p-value
Survey complete	0.94	0.93	0.000	0.97
All dead	0.01	0.01	0.001	0.87
Merged with another	0.00	0.00	-0.001	0.75
Refused	0.02	0.03	0.009	0.15
House not found	0.00	0.00	-0.003	0.24
Migrated	0.02	0.02	-0.006	0.30

Notes. Each row in the table corresponds to a survey status (survey completed or reason for non-completion). The proportions are relative to all households surveyed in round 1. The first two columns are the proportion of tenable and untenable households with that survey status in round 2. The third column is the difference between tenable and untenable proportions estimated with a slum fixed effect. The fourth column gives us the p-value of the difference being different from 0.

Appendix D: Balance tables

Balance on baseline covariates by land tenability, in the full sample frame, in the round-1 survey sample and in the round-2 panel.

Table D1: Balance table: Full sample frame

	Untenable		Tenable				
	n	mean	n	mean	Diff	Confidence interval	p-value
Occupied area (sqft)	16531	508.93	19317	509.58	-6.016	-15.898 - 3.865	0.23
Area per person (sqft)	12811	196.03	15581	198.36	-0.723	-6.987 - 5.540	0.82
Annual Income (winsorized)	14380	60618.89	17136	59895.58	789.965	-654.940 - 2,234.869	0.28
Income per capita (Rs)	12798	23164.28	15581	23168.80	537.454	-780.461 - 1,855.369	0.42
Permanent roof	14416	0.11	17122	0.10	0.005	-0.003 - 0.013	0.23
Brick wall	14422	0.60	17160	0.63	0.001	-0.012 - 0.013	0.92
Toilet in house	14421	0.42	17158	0.34	0.000	-0.012 - 0.012	0.97
Direct access to road	14421	0.75	17160	0.69	-0.006	-0.017 - 0.004	0.23
HH size	12823	3.24	15588	3.29	0.009	-0.037 - 0.055	0.70
Number of children	12816	0.91	15625	0.91	-0.028	-0.063 - 0.007	0.11
Average household age	12799	33.86	15585	33.63	0.015	-0.439 - 0.469	0.95
Proportion female	12822	0.51	15588	0.51	-0.008	-0.016 - -0.001	0.03*

Notes. The sample includes all households in the 11 cities that are a part of the sample. The first 4 columns present the mean and number of observations for individuals in tenable and untenable lands, respectively. The last three columns present the coefficient of regressing the outcome on the tenability of the land with slum fixed effects, the 95% confidence interval, and the p-value. We can reject differences between the two groups larger than the confidence intervals. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Table D2: Balance table: round-1 survey sample

	Untenable		Tenable				
	n	mean	n	mean	Diff	Confidence interval	p-value
Plot chars							
Occupied area (sqft)	1814	497.13	4103	507.41	16.989	-7.624 - 41.602	0.18
Area per person (sqft)	1814	189.71	4103	190.86	-1.661	-14.334 - 11.011	0.80
Income							
Annual income (Rs.)	1814	58878.10	4103	58852.27	-73.788	-3,787.355 - 3,639.780	0.97
Income per capita (Rs)	1814	20850.15	4103	21840.96	-651.480	-2,295.642 - 992.683	0.44
House quality							
Concrete roof	1814	0.06	4103	0.10	-0.003	-0.024 - 0.017	0.75
Concrete/tiled roof	1814	0.30	4103	0.29	-0.002	-0.033 - 0.029	0.90
Brick wall	1814	0.53	4103	0.58	0.014	-0.022 - 0.049	0.45
Toilet in house	1814	0.33	4103	0.33	0.008	-0.025 - 0.040	0.64
Direct access to road	1814	0.75	4103	0.68	0.022	-0.006 - 0.050	0.12
Demographics							
HH size	1814	3.29	4103	3.31	0.108	-0.001 - 0.217	0.05
Number of children	1814	0.93	4103	0.94	0.088	0.002 - 0.174	0.04*
Average household age	1814	33.39	4103	33.38	-0.971	-2.088 - 0.145	0.09
Proportion female	1814	0.51	4103	0.50	-0.017	-0.036 - 0.002	0.08
Education							
Highest adult edu	1704	6.68	3877	6.91	0.083	-0.298 - 0.463	0.67
Mean adult edu	1692	4.95	3858	5.16	0.113	-0.204 - 0.430	0.48
Migration							
Years in slum	1798	58.91	4066	54.19	0.797	-1.466 - 3.060	0.49

Notes. The sample is all households surveyed in the round-1 endline. Baseline covariates are measured in the pre-intervention survey. The first four columns give the number of observations and mean for untenable and tenable households; the last three give the difference (coefficient on the tenability indicator with a slum fixed effect), its 95% confidence interval, and the p-value. Income is the raw annual figure in rupees. Covariates match Table 2.
$^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$

Appendix E: Effects on all house quality measures

Tables E1–E2 report the ITT effect of tenability on every individual house-quality measure. All specifications are ITT with slum fixed effects and baseline controls.

Table E1: Short-run ITT effects on all house-quality measures

	Untenable	Tenable	N	ITT	Confidence interval	p-value
Roof						
Concrete	0.14	0.25	5917	0.058	0.029 - 0.088	0.00***
Tiled	0.30	0.37	5917	0.038	0.003 - 0.073	0.03*
Good asbestos	0.71	0.72	5917	-0.011	-0.046 - 0.023	0.51
Bad asbestos	0.97	0.96	5917	0.005	-0.009 - 0.019	0.50
Wall						
Brick	0.66	0.69	5917	0.022	-0.014 - 0.057	0.23
Mud	0.86	0.87	5917	0.026	-0.002 - 0.054	0.07
Metal/asbestos	0.97	0.96	5917	0.009	-0.006 - 0.025	0.24
Painted walls	0.55	0.56	5917	0.012	-0.024 - 0.048	0.52
Floor						
Tiles	0.05	0.07	5917	0.012	-0.006 - 0.030	0.19
Cement	0.71	0.75	5917	-0.000	-0.033 - 0.033	0.99
Toilet						
In-house w water	0.15	0.18	5917	-0.000	-0.028 - 0.028	1.00
In house	0.33	0.33	5917	0.000	-0.033 - 0.034	0.98
Exclusive	0.52	0.57	5917	0.018	-0.015 - 0.052	0.29
Shared	0.57	0.60	5917	0.024	-0.009 - 0.058	0.15
Other features						
Door with latch	0.86	0.86	5917	0.012	-0.016 - 0.039	0.40
Concealed wires	0.39	0.45	5917	0.019	-0.017 - 0.055	0.31
Num windows	1.35	1.57	5910	0.022	-0.098 - 0.142	0.72
Num rooms	2.74	2.75	5911	-0.047	-0.149 - 0.054	0.36
Theft threat	0.18	0.17	5917	-0.011	-0.039 - 0.018	0.47
Problems during rain						
Roof not leaking	0.27	0.30	5917	0.024	-0.012 - 0.059	0.19
Door not leaking	0.52	0.49	5917	0.005	-0.034 - 0.043	0.81
Wall not leaking	0.51	0.49	5917	-0.009	-0.048 - 0.029	0.64
No water logging	0.85	0.84	5917	0.031	0.004 - 0.059	0.03*

Notes. This table presents ITT estimates of living on tenable lands on all house-quality measures recorded in the first survey round in 2023. Each row is a separate regression of the measure on the tenability indicator, estimated using specification 3 with slum fixed effects and baseline controls. The first two columns are the means for untenable and tenable households, and the third column is the number of observations. The last three columns present the ITT coefficient, its 95% confidence interval, and the p-value, computed using robust standard errors. All measures are indicators except the number of windows and rooms. Asbestos roofs are classified as good or bad by the enumerator. Theft threat indicates that the household reports its house is at risk of theft. The measures under "Problems during rain" are coded so that higher values are better: the roof, door and walls did *not* leak and there was *no* waterlogging during the last rainy season. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table E2: Medium-run ITT effects on all house-quality measures

	Untenable	Tenable	N	ITT	Confidence interval	p-value
Roof						
Concrete	0.18	0.35	6001	0.096	0.067 - 0.125	0.00***
Tiled	0.28	0.42	6001	0.082	0.050 - 0.113	0.00***
Wall						
Brick	0.76	0.82	6001	0.021	-0.010 - 0.052	0.18
Mud	0.98	0.99	6001	0.008	-0.003 - 0.019	0.13
Painted walls	0.61	0.64	6001	0.017	-0.017 - 0.051	0.33
Floor						
Tiles	0.08	0.12	6001	0.023	0.001 - 0.044	0.04*
Cement	0.76	0.82	6001	0.027	-0.003 - 0.057	0.07
Covered	0.76	0.82	6001	0.027	-0.003 - 0.057	0.07
Toilet						
In-house w water	0.17	0.19	6001	0.003	-0.025 - 0.030	0.86
In house	0.24	0.26	6001	0.004	-0.027 - 0.035	0.80
Exclusive	0.60	0.66	6001	0.007	-0.023 - 0.038	0.64
Shared	0.66	0.70	6001	0.004	-0.026 - 0.033	0.81
Any access	0.66	0.70	6001	0.004	-0.026 - 0.033	0.81
Other features						
Door with latch	0.91	0.91	6001	0.004	-0.020 - 0.027	0.76
Concealed wires	0.44	0.51	6001	-0.001	-0.036 - 0.034	0.96
Num windows	1.52	1.75	5913	0.030	-0.079 - 0.139	0.59
Num rooms	2.62	2.72	5916	-0.098	-0.191 - -0.005	0.04*
Problems during rain						
Roof not leaking	0.23	0.32	5829	0.040	0.006 - 0.074	0.02*
Door not leaking	0.45	0.47	5829	0.010	-0.029 - 0.048	0.62
No water logging	0.80	0.83	5829	-0.005	-0.036 - 0.025	0.73

Notes. As in Table E1, for the second survey round in 2026. Each row is a separate regression of the measure on the tenability indicator, estimated using specification 3 with slum fixed effects and baseline controls, with robust standard errors. $^{*}\,p < 0.05$, $^{**}\,p < 0.01$, $^{***}\,p < 0.001$.

Table E3: ITT on alternative house-quality index definitions

Panel A: Short-run effects (round 1)

	(1) PCA index (endline)	(2) PCA index (BL weights)	(3) Rent predictors	(4) Satisfaction predictors
Tenable	0.0637	0.142**	0.0556	0.0447
	(0.0378)	(0.0434)	(0.0409)	(0.0392)
Untenable mean	0.00	0.37	-0.00	0.00
Slum FE & BL controls	Yes	Yes	Yes	Yes
N	5910	5917	5910	5910

Panel B: Medium-run effects (round 2)

	(1) PCA index (endline)	(2) PCA index (BL weights)	(3) Rent predictors	(4) Satisfaction predictors
Tenable	0.0881*	0.221***	0.0544	0.0821*
	(0.0352)	(0.0402)	(0.0356)	(0.0379)
Untenable mean	-0.00	0.57	-0.00	0.00
Slum FE & BL controls	Yes	Yes	Yes	Yes
N	5744	6001	5744	5744

Notes. This table presents ITT estimates of living on tenable lands on four alternative definitions of the house quality index. Column 1 is the first principal component of the house-quality measures, with weights estimated on the endline data of that round. Column 2 applies the principal-component weights estimated on the 2017 baseline census to the roof, wall and toilet features recorded in both the baseline and the endline, and is standardized relative to untenable households at baseline; its untenable mean therefore reflects the improvement in house quality since 2017. Column 3 weights each house-quality measure by its coefficient in a regression of the monthly rent the household reports it would be willing to pay for its current house on all house-quality measures. Column 4 weights each measure by its coefficient in a regression of the household's self-reported satisfaction with its house on the same measures. Columns 1, 3 and 4 are standardized relative to untenable households in that round. The house-quality measures are those listed in Tables E1 and E2. The estimates are coefficients from specification 3. All regressions include slum fixed effects and baseline controls. Robust standard errors in parentheses. Panel A presents results from the first round in 2023 and Panel B presents results from the second round in 2026. $^{*}\,p < 0.05$, $^{**}\,p < 0.01$, $^{***}\,p < 0.001$.

Appendix F: ToT estimates for PMAY using IV regressions

Table F1: IV estimates: house quality

Panel A: Short-run effects

	House quality index	House quality components				
	(1)	(2)	(3)	(4)	(5)	(6)
		Roof	Wall	Floor	Toilet	Concealed wires
Received PMAY	0.649	0.596	0.223	-0.00121	0.250	0.193
	(0.382)	(.)	(0.182)	(0.170)	(0.172)	(0.185)
Untenable mean	2.9e-10	0.14	0.66	0.71	0.57	0.39
Tenable mean	0.12	0.25	0.69	0.75	0.60	0.45
Slum FE & BL controls	Yes	Yes	Yes	Yes	Yes	Yes
N	5910	5917	5917	5917	5917	5917

Panel B: Medium-run effects

	House quality index	House quality components				
	(1)	(2)	(3)	(4)	(5)	(6)
		Roof	Wall	Floor	Toilet	Concealed wires
Received PMAY	0.750**	0.801***	0.174	0.228	0.0296	-0.00689
	(0.289)	(0.122)	(0.127)	(0.124)	(0.124)	(0.146)
Untenable mean	-1.7e-09	0.18	0.76	0.76	0.66	0.44
Tenable mean	0.25	0.35	0.82	0.82	0.70	0.51
Slum FE & BL controls	Yes	Yes	Yes	Yes	Yes	Yes
N	5744	6001	6001	6001	6001	6001

This table presents IV estimates of the effect of receiving the PMAY grant on the house quality index and some components of the index. Grant receipt is instrumented by living on tenable land. The dependent variables are the house quality index defined as the principal component of all individual measures, and indicators for whether the house has a concrete roof, brick wall, covered floor, access to any toilet, and concealed wires which are safer. All regressions include slum fixed effects and baseline controls. Untenable and tenable means are the means of the dependent variable among untenable and tenable households in the estimation sample. Robust standard errors in parentheses. Panel A presents results from the first round in 2023 and Panel B presents results from the second round in 2026. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table F2: IV estimates: adaptation to heat

	Outside − inside temperature		
	(1) All days	(2) Hot days	(3) Not-hot days
Received PMAY	0.642	1.867	0.622
	(0.450)	(1.006)	(0.548)
Untenable mean	1.53	1.59	1.49
Tenable mean	1.63	1.73	1.59
Slum FE & BL controls	Yes	Yes	Yes
N	5903	1851	4052

This table presents IV estimates of the effect of receiving the PMAY grant on the difference between the temperature outside and the temperature inside the house (℃). Grant receipt is instrumented by living on tenable land. Column 1 includes all survey days, column 2 restricts to households surveyed on days when the outside temperature was above 32℃, and column 3 to the remaining days. All regressions include slum fixed effects and baseline controls. Untenable and tenable means are the means of the dependent variable among untenable and tenable households in the estimation sample. Robust standard errors in parentheses. Temperature was measured only in the second round in 2026. $^{*}\, p < 0.05$, $^{**}\, p < 0.01$, $^{***}\, p < 0.001$.

Table F3: IV estimates: health

Panel A: Short-run effects

	GAD-7	U5MR	Fever	Diarrhoea	Unsatisfied with life	Unsatisfied with sleep
	(1)	(2)	(3)	(4)	(5)	(6)
Received PMAY	2.349	-0.593	0.0497	0.0264	0.0753	-0.256
	(2.279)	(0.343)	(0.126)	(.)	(0.544)	(0.518)
Untenable mean	6.56	0.22	0.24	0.082	1.04	0.70
Tenable mean	7.47	0.21	0.26	0.082	1.08	0.78
Slum FE & BL controls	Yes	Yes	Yes	Yes	Yes	Yes
N	5875	1630	5917	3731	5900	5902

Panel B: Medium-run effects

	U5MR	Fever	Diarrhoea	Unsatisfied with life	Unsatisfied with sleep
	(1)	(2)	(3)	(4)	(5)
Received PMAY	-0.129	0.0408	0.278*	-0.691	0.0686
	(0.256)	(0.112)	(0.110)	(0.444)	(0.459)
Untenable mean	0.18	0.31	0.11	1.26	1.09
Tenable mean	0.19	0.32	0.12	1.20	1.07
Slum FE & BL controls	Yes	Yes	Yes	Yes	Yes
N	1358	6000	3599	5986	5975

This table presents IV estimates of the effect of receiving the PMAY grant on health outcomes. Grant receipt is instrumented by living on tenable land. The dependent variables are the GAD-7 anxiety score (Panel A only), the under-5 mortality rate, the share of household members who had a fever in the last month, the share of household members who had diarrhoea in the last month, and dissatisfaction with life and with sleep (reverse-coded so that higher values indicate greater dissatisfaction). The GAD-7 module was not collected in the second round. All regressions include slum fixed effects and baseline controls. Untenable and tenable means are the means of the dependent variable among untenable and tenable households in the estimation sample. Robust standard errors in parentheses. Panel A presents results from the first round in 2023 and Panel B presents results from the second round in 2026. $^{*}\ p < 0.05$, $^{**}\ p < 0.01$, $^{***}\ p < 0.001$.

Table F4: IV estimates: income

Panel A: Short-run effects

	Total income	Business income	Wage income	Any business
	(1)	(2)	(3)	(4)
Received PMAY	1278.4	854.8	-29.09	0.126
	(1466.8)	(693.8)	(1187.3)	(0.183)
Untenable mean	3228.3	576.4	2142.8	0.32
Tenable mean	3423.2	621.5	2226.2	0.33
Slum FE & BL controls	Yes	Yes	Yes	Yes
N	5897	5917	5917	5910

Panel B: Medium-run effects

	Total income	Business income	Wage income	Any business
	(1)	(2)	(3)	(4)
Received PMAY	20.41	-561.8	437.5	-0.0389
	(1428.6)	(678.8)	(1484.6)	(0.145)
Untenable mean	3295.8	370.4	3175.8	0.29
Tenable mean	3680.0	628.1	3369.3	0.31
Slum FE & BL controls	Yes	Yes	Yes	Yes
N	6000	6000	4384	6000

This table presents IV estimates of the effect of receiving the PMAY grant on income. Grant receipt is instrumented by living on tenable land. The dependent variables are total household monthly income, business income, and wage income, each divided by the number of household members (Rs), and whether any household member runs a business. All regressions include slum fixed effects and baseline controls. Untenable and tenable means are the means of the dependent variable among untenable and tenable households in the estimation sample. Robust standard errors in parentheses. Panel A presents results from the first round in 2023 and Panel B presents results from the second round in 2026. $^{*}\, p < 0.05$, $^{**}\, p < 0.01$, $^{***}\, p < 0.001$.

Table F5: IV estimates: assets

Panel A: Short-run effects

	Asset value	TV	Fridge	Bed	Smartphone	Chair
	(1)	(2)	(3)	(4)	(5)	(6)
Received PMAY	1672.3	-0.0202	-0.157	-0.255	0.333	-0.277
	(9515.4)	(0.207)	(0.194)	(0.407)	(0.421)	(0.808)
Untenable mean	36686.8	0.70	0.35	1.16	1.23	2.15
Tenable mean	37371.6	0.69	0.35	1.15	1.25	2.26
Slum FE & BL controls	Yes	Yes	Yes	Yes	Yes	Yes
N	5903	5912	5907	5912	5909	5910

Panel B: Medium-run effects

	Asset value	TV	Fridge	Bed	Smartphone	Chair
	(1)	(2)	(3)	(4)	(5)	(6)
Received PMAY	-9730.4	-0.0208	0.0171	-0.287	-0.348	0.0564
	(9720.1)	(0.192)	(0.179)	(1.016)	(0.353)	(0.566)
Untenable mean	54325.9	0.55	0.39	3.08	1.56	2.10
Tenable mean	52360.1	0.57	0.40	2.67	1.56	2.23
Slum FE & BL controls	Yes	Yes	Yes	Yes	Yes	Yes
N	5993	5999	5999	5995	5999	5998

This table presents IV estimates of the effect of receiving the PMAY grant on household assets. Grant receipt is instrumented by living on tenable land. The dependent variables are the total value of assets in the house (Rs) and the number of televisions, refrigerators, beds, smartphones, and chairs owned by the household. All regressions include slum fixed effects and baseline controls. Untenable and tenable means are the means of the dependent variable among untenable and tenable households in the estimation sample. Robust standard errors in parentheses. Panel A presents results from the first round in 2023 and Panel B presents results from the second round in 2026. $^{*}\ p < 0.05$, $^{**}\ p < 0.01$, $^{***}\ p < 0.001$.

Appendix G: Proofs of propositions

Proposition 1. *The probability p_i that a household with given baseline characteristics takes up the program is given by*

$$p_i = \frac{e^{u_{yi}/\beta}}{e^{u_{yi}/\beta} + e^{u_{ni}/\beta}} = \frac{1}{1 + e^{(u_{ni}-u_{yi})/\beta}} \tag{11}$$

Proof. The household takes up the grant only if the utility gained is positive.

$$p_i = \Pr(\lambda_i > u_{ni} - u_{yi})$$

Since λ_i has a logistic distribution, the cumulative distribution function for λ_i is given by

$$\Pr(\lambda_i < x) = \frac{1}{1 + e^{-x/\beta}}$$

Hence

$$p_i = 1 - \frac{1}{1 + e^{(u_{yi}-u_{ni})/\beta}} = \frac{e^{(u_{yi}-u_{ni})/\beta}}{1 + e^{(u_{yi}-u_{ni})/\beta}} = \frac{1}{1 + e^{(u_{ni}-u_{yi})/\beta}}$$

■

Proposition 2. *Under assumption 1, the expected welfare transferred to a household i is given by*

$$E[W_i] = \beta \log\left(\frac{1}{1 - p_i}\right) \tag{12}$$

Proof. For a given λ_i,

$$W_i = \max\left(u_{yi} + \lambda_i, u_{ni}\right)$$

Taking expectations over different values of λ_i

$$
\begin{aligned}
E[W_i] &= \int_{\mathbb{R}} \max(u_{yi} + \lambda_i - u_{ni}, 0) pdf(\lambda_i) d\lambda_i \\
&= \int_{-(u_{yi}-u_{ni})}^{\infty} (u_{yi} - u_{ni} + \lambda_i) \frac{e^{-\lambda_i/\beta}}{\beta(1+e^{-\lambda_i/\beta})^2} d\lambda_i \\
&= (u_{yi} - u_{ni}) \left(1 - \frac{1}{1+e^{(u_{yi}-u_{ni})/\beta}}\right) + \int_{-(u_{yi}-u_{ni})}^{\infty} \lambda_i \frac{e^{-\lambda_i/\beta}}{\beta(1+e^{-\lambda_i/\beta})^2} d\lambda_i \\
&= \beta \log\left(\frac{p_i}{1-p_i}\right) p_i + \int_{-(u_{yi}-u_{ni})/\beta}^{\infty} \beta t \frac{e^{-t}}{(1+e^{-t})^2} dt \\
&= \beta \log\left(\frac{p_i}{1-p_i}\right) p_i + \beta \left[\frac{t}{1+e^{-t}} - \log(1+e^t)\right]_{-(u_{yi}-u_{ni})/\beta}^{\infty} \\
&= \beta \log\left(\frac{p_i}{1-p_i}\right) p_i + \beta \log\left(e^{-(u_{yi}-u_{ni})/\beta} + 1\right) - \frac{-(u_{yi}-u_{ni}) e^{-(u_{yi}-u_{ni})/\beta}}{(1+e^{-(u_{yi}-u_{ni})/\beta})} \\
&= \beta \log\left(\frac{p_i}{1-p_i}\right) p_i + \beta \log 1/p_i + \beta \log\left(\frac{p_i}{1-p_i}\right)(1-p_i) \\
&= \beta \log\left(\frac{1}{1-p_i}\right)
\end{aligned}
$$

■

Appendix H: Welfare under a normal preference shock

In this section I recompute the welfare curve in Section 5 using a different assumption.

In assumption 1 we assumed that λ_i has a logistic$(0, \beta)$ distribution. Suppose instead that it has a normal distribution.

Assumption 1n. *λ_i is distributed as $N(0, \beta^2)$*

Then the expressions for take-up probability and welfare would change slightly.

Proposition 1n. *The probability p_i that a household with given baseline characteristics takes up the program is given by*

$$p_i = 1 - \Phi((u_{ni} - u_{yi})/\beta) \tag{13}$$

Proposition 2n. *Under the modified Assumption 1n, the expected welfare transferred to a household i is given by*

$$E[W_i] = -\beta\Phi^{-1}(1 - p_i)p_i + \beta\phi(\Phi^{-1}(1 - p_i)) \tag{14}$$

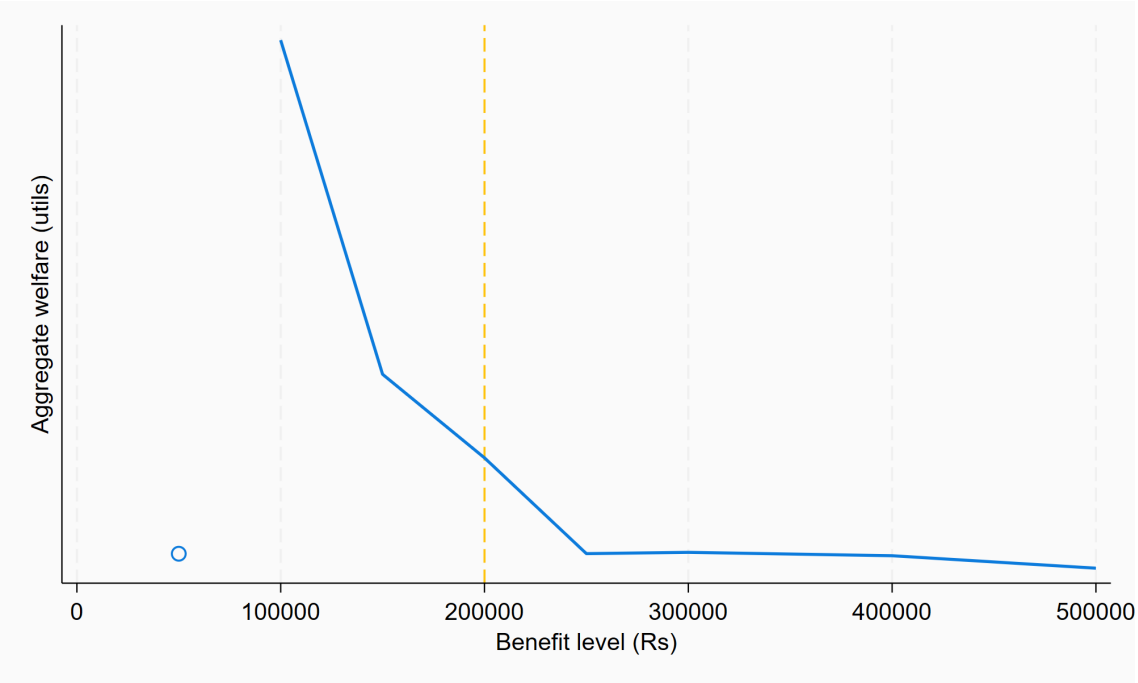
Proof. Let $\Delta_i = u_{yi} - u_{ni}$ and let $\phi_\beta(x) = \beta^{-1}\phi(x/\beta)$ denote the $N(0, \beta^2)$ density. From Equation 13, $\Phi^{-1}(1 - p_i) = -\Delta_i/\beta$. For a given λ_i, $W_i = \max(\Delta_i + \lambda_i, 0)$, so

$$\begin{aligned}
E[W_i] &= \int_{-\Delta_i}^{\infty} (\Delta_i + \lambda_i)\, \phi_\beta(\lambda_i)\, d\lambda_i \\
&= \Delta_i\, (1 - \Phi(-\Delta_i/\beta)) + \int_{-\Delta_i}^{\infty} \lambda_i\, \phi_\beta(\lambda_i)\, d\lambda_i \\
&= \Delta_i\, p_i + \beta \int_{-\Delta_i/\beta}^{\infty} t\, \phi(t)\, dt \\
&= \Delta_i\, p_i + \beta\, [-\phi(t)]_{-\Delta_i/\beta}^{\infty} \\
&= \Delta_i\, p_i + \beta\, \phi(-\Delta_i/\beta) \\
&= -\beta\, \Phi^{-1}(1 - p_i)\, p_i + \beta\, \phi\big(\Phi^{-1}(1 - p_i)\big)
\end{aligned}$$

■

Figure H1 recomputes the welfare curve in Figure 6 under Assumption 1n, holding fixed the take-up cells, the budget and the lottery rule from Section 5.3. Because β scales every household's expected welfare by the same factor, the shape of the curve and the welfare-maximizing grant do not depend on the value of β.

Figure H1: Welfare gain at different grant sizes: normal preference shock



Notes. This figure presents estimated aggregate welfare transferred at different benefit levels when the idiosyncratic preference shock is distributed $N(0, \beta^2)$ instead of logistic. Take-up probabilities are cell means of stated take-up from the round-2 survey, with cells defined by baseline income quintile and baseline house-quality quintile. The budget is fixed at half the current level and grants are rationed by lottery when demand exceeds the number of grants. The hollow marker at Rs 50,000 is a corner case in which take-up is low and part of the budget goes unspent. The dashed line marks the current benefit of Rs 200,000.

The curve in Figure H1 closely tracks Figure 6 up to a grant of Rs 250,000. At higher grant sizes it no longer rises, which strengthens the case against a larger grant. The qualitative conclusion that reducing the grant size increases aggregate welfare transferred remains the same.